TrekNepal - Artifact Design and Test Plan Documentation

Student Id : 2431868

Student Name : Prakriti Khadka

Section : L6CG24

Reader : Uttam Acharya

Supervisor : Priyanka Aryal

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1. Introduction to TrekNepal Artifact

TrekNepal is a web-based tourism platform designed to help trekkers explore trekking destinations in Nepal, book trekking packages, access route information, guides, and travel assistance. (Board, 2024)The artifact represents the full functioning TrekNepal web application that focuses on improving accessibility, planning, and safety for trekkers. (Maxim, 2020)

Aim:

- To develop a comprehensive web-based platform that helps trekkers explore, plan, and book trekking experiences in Nepal.

Objectives:

- Provide a user-friendly interface for exploring trekking destinations and packages.
- Streamline the booking process for treks and guides.
- Enhance safety by providing route details, elevation profiles, and weather updates.
- Integrate real-time data for maps, routing, and travel assistance.

2. System Architecture Overview

TrekNepal follows a three-tier architecture consisting of:

• Presentation Layer (Web UI)

• Application Layer (Backend Services and APIs)

• Data Layer (Database and Storage)

The architecture ensures scalability, security, and efficiency in user interaction and data handling.

3. Functional Modules

The system is divided into key modules including:

• User Registration & Authentication

• Trek Package Browsing & Booking System

• Guide Management

• Trek Details

• Reviews & Ratings

• Admin Dashboard

• Payment Integration Module

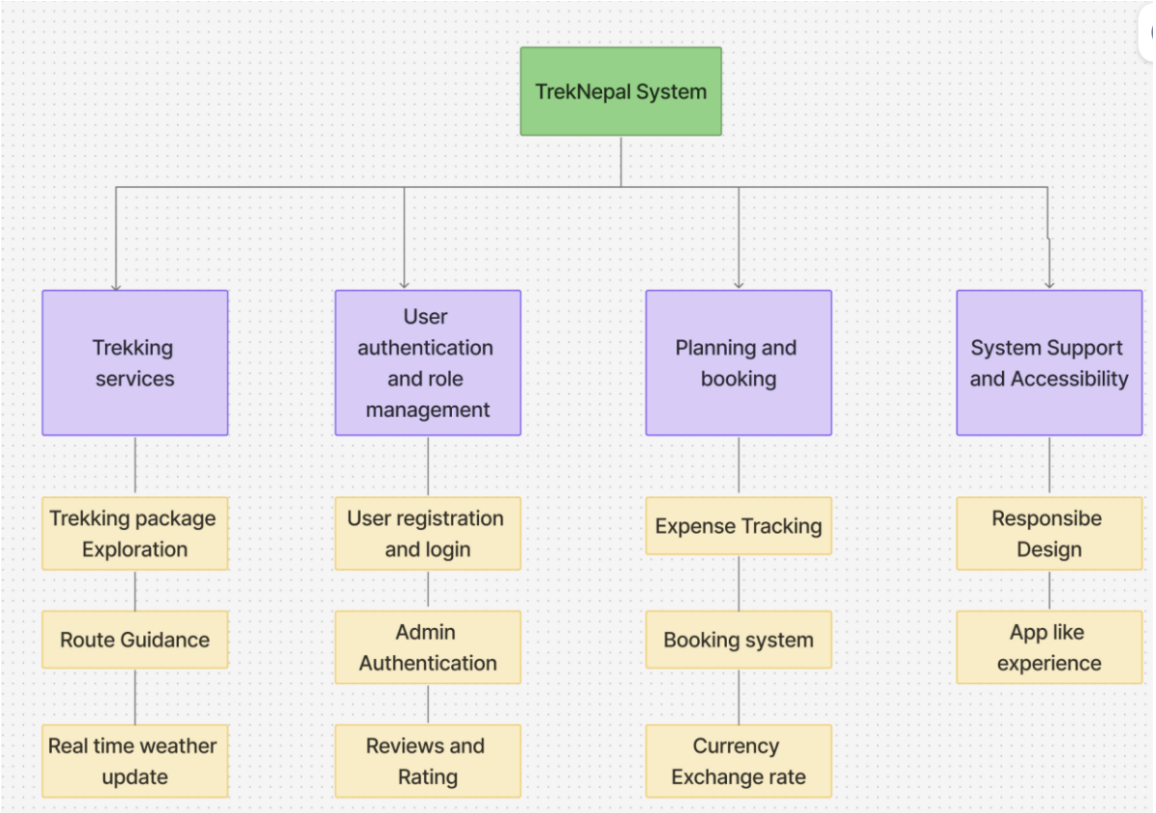

Figure 1:Functional Decomposition

4. Design Diagrams

Users interact with the system for registration, browsing packages, booking, payment, and Reviews. Admin manages system data, packages, guide and users.

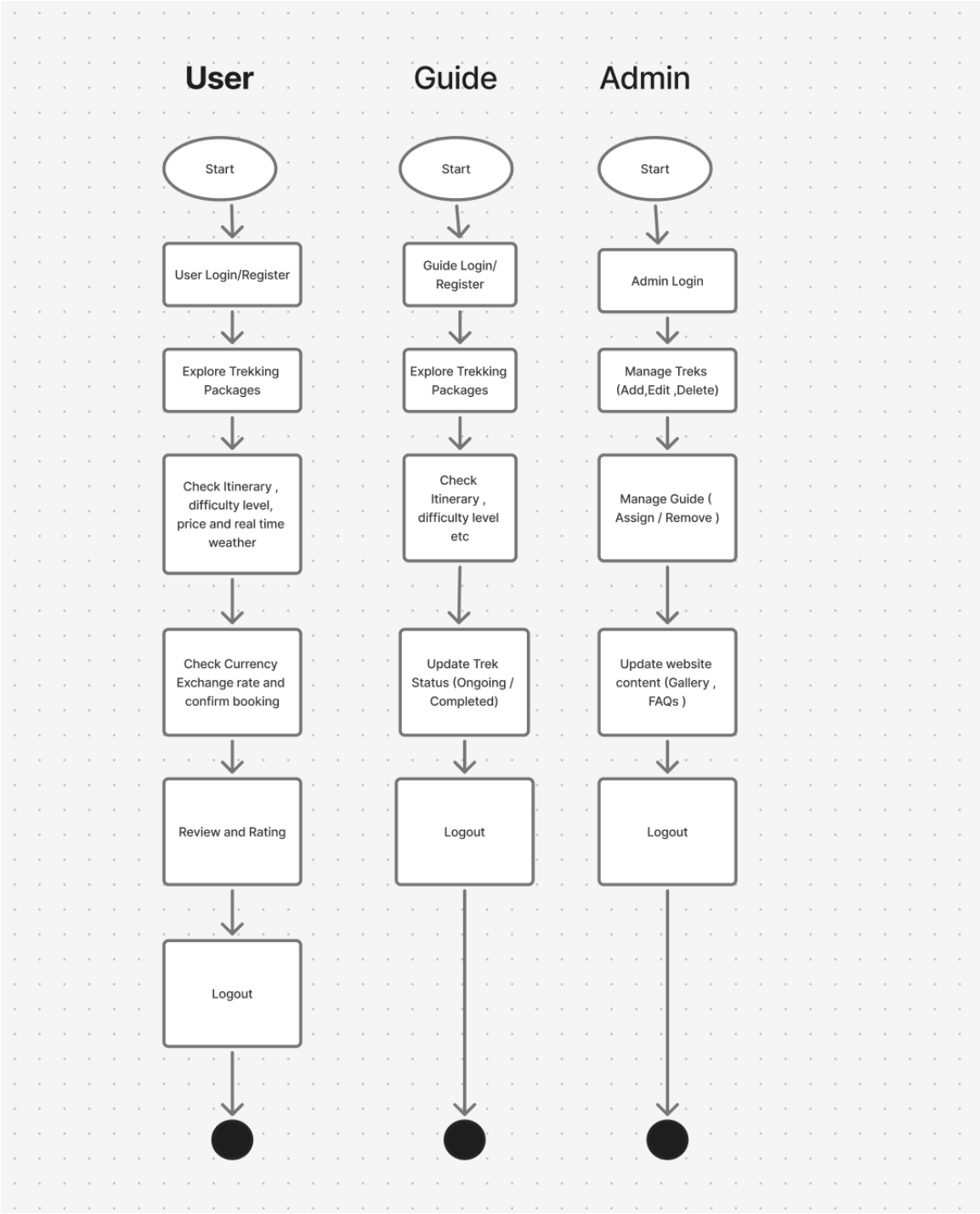

Figure 2:Flow Chart Diagram

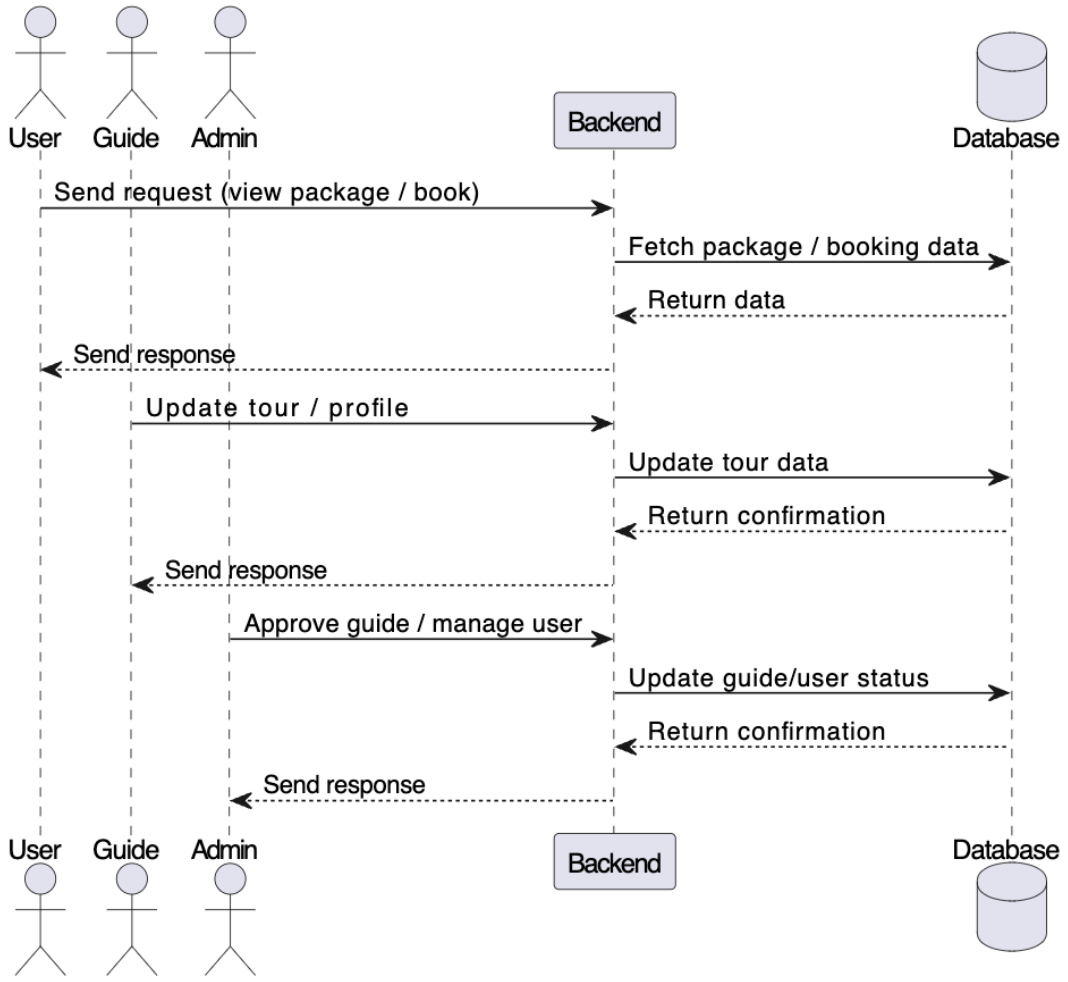

Figure 3:Sequence Diagram

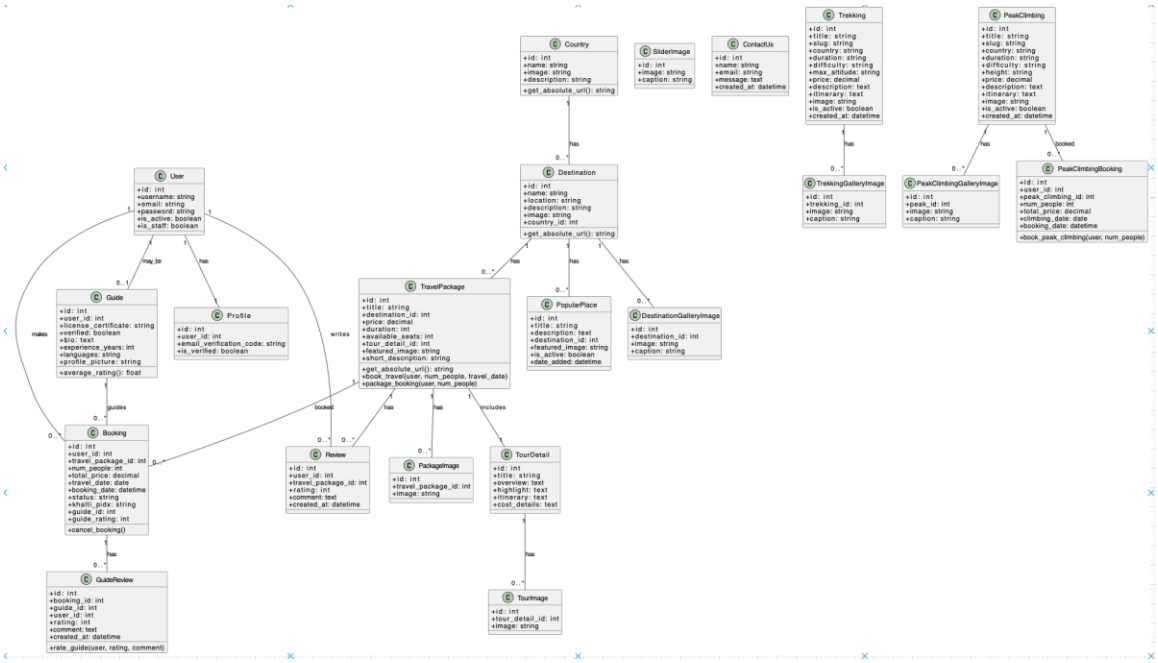

Figure 4:Class Diagram

4.1 Use Case: Interactive Map Integration

4.1.1 Use Case Details

This use case explains how the **interactive trek planning map** is integrated into the TrekNepal system.After logging in, the user can open the Trek Map page to plan trekking routes by entering a start point, destination, and optional waypoints.The system loads the map, geocodes locations, displays markers, calculates the route, and shows elevation profiles. ((Project OSRM, 2024)

Actor(s)

Registered User (Logged-in User)

4.1.2 Detailed Process Flow

When the user opens the Home page, this use case begins.

Interaction	**Response**
User logs into TrekNepal system	System authenticates user and opens dashboard
User clicks on to Map button	System loads the Trek Map interface
User clicks Allow	The system shows the user's current location on the map

User enters start point & destination	System geocodes the entered locations
User adds optional waypoints	System lists and displays waypoint inputs
User clicks Get Trek Route	System fetches the walking/trekking route and draws it on the map
System fetches elevation data	System displays elevation chart + ascent/descent + distance/time
User zooms or pans the map	Map updates smoothly without page reload
User clicks Download GPX/KML	System prepares trek route export file

4.1.3 Special Requirements

Special Requirements
● Internet connection is required for map tiles, geocoding, routing, and elevation APIs. ● All entered locations must be valid and geocodable. ● User's GPS permission must be granted to show "current location." ● OSRM routing API must be online to compute trek routes. ● Elevation API must be available to compute elevation profile.

4.1.4 Pre-conditions

• User must be logged in to TrekNepal.

• User must open the **Trek Map** page.

• Map API (MapLibre) must be connected and working.

• OSRM routing API must be available. (USGS, 2024)

• Elevation API must be available.

4.1.5 Post-conditions

• User can see the full trek route on the map.

• User can view markers for start, destination, and waypoints.

• User can view elevation profile, ascent, descent, distance, and estimated time.

• User can share trek link through download GPX/KML.

4.1.6 Keyword

Interactive Trek Map

4.1.7 Sample User Interface

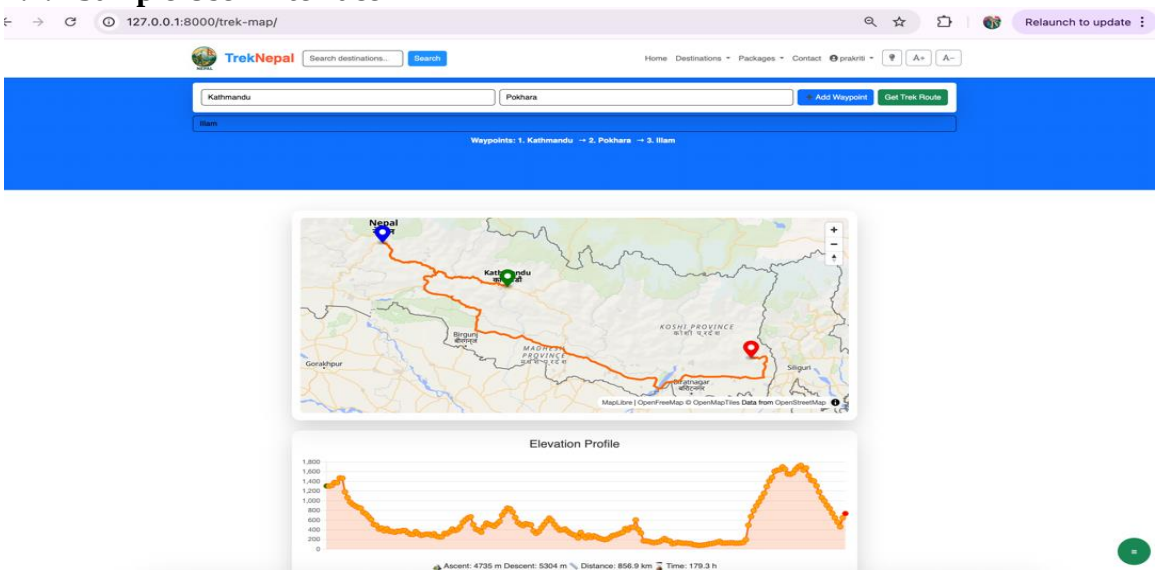

Figure 5:Sample User Interface for Interactive Trek Map

4.1.8 Property Sheet

S.N	Field Name	Basic Type	Data Type	Field Size	Validation?	Enable?

1	Map Area	Map Component	–	–	Must load correctly	Yes
2	Start/Destination Input	Text	String	–	Must be valid place name	Yes
3	Waypoint Input	Text	String	–	Optional	Yes
4	Place Marker	Icon Marker	–	–	Valid coordinates	Yes
5	Directions / Route Line	polyline	-	-	API must return route	Yes
6	GPX/KML Download Buttons	Button	-	-	Enabled after route is generated	Yes

4.1.9 Process Flow Diagram

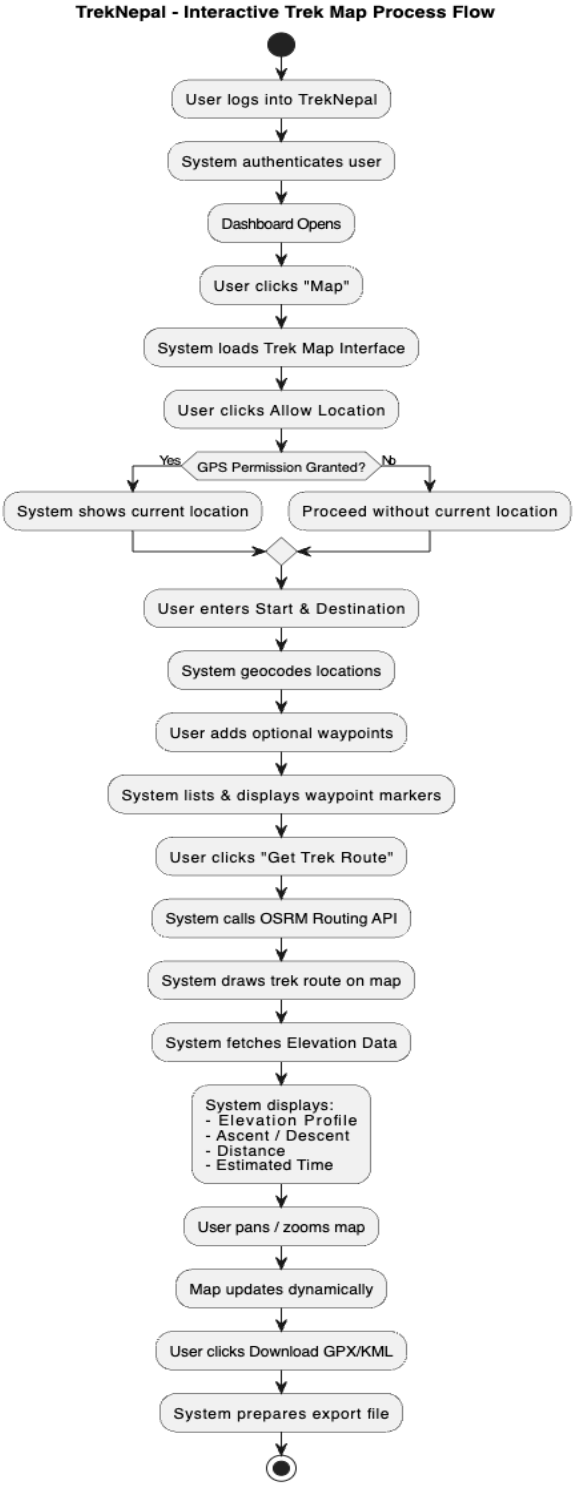

Figure 6:Process Flow Diagram for Interactive Trek Map

4.1.10 Use-Case Diagram

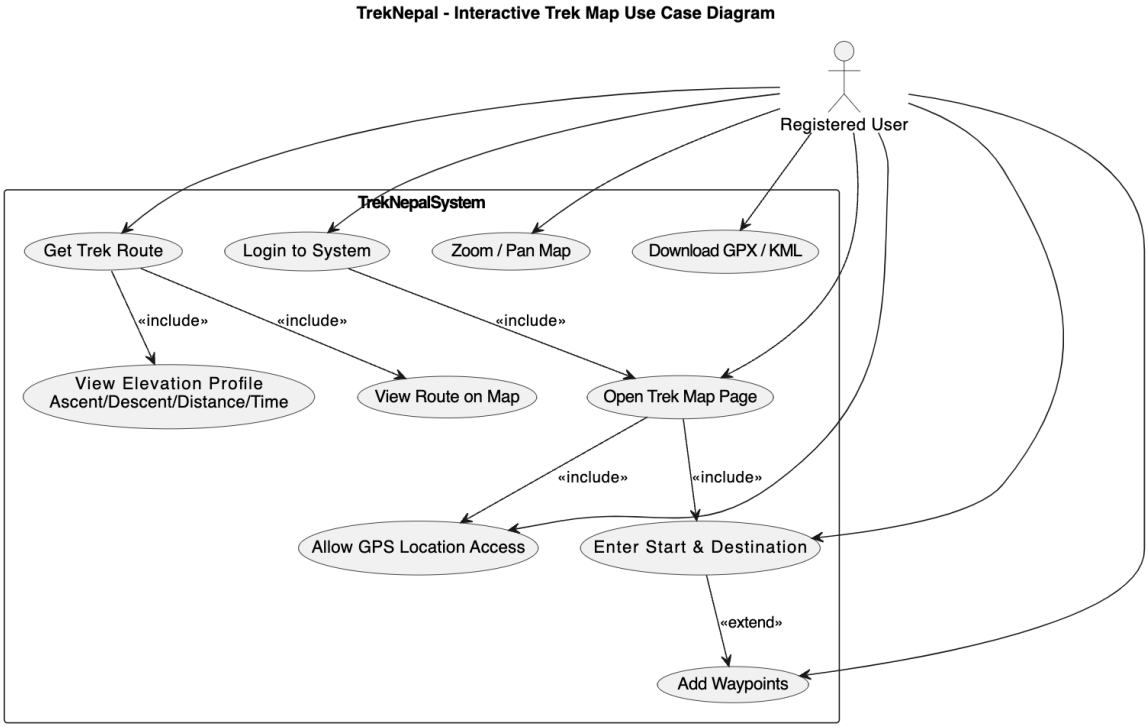

Figure 7:Use case Diagram of Interactive Trek Map

4.2 Use Case: Booking System

4.2.1 Use Case Details

This use case describes how the booking system functions within the TrekNepal platform. After logging in, the user accesses the home page, where they can explore available trekking packages. The user selects a package, enters the number of participants and preferred date, and proceeds to payment. The booking is confirmed only after the payment is successfully completed, after which the system finalizes the reservation. (Wallet, 2024)

Actor(s)

Registered User (Logged-in User)

4.2.2 Detailed Process Flow

When the user opens the Home page, this use case begins.

Interaction	**Response**
User logs into TrekNepal system	System authenticates user and opens dashboard

User clicks on to Explore Pacakages Button	System loads the available packages
User clicks the input fields	The system shows the button of proceed payment
User clicks the proceed payment button	System redirect user to the payment template
User proceed the payment by entering the amount	System work on it by matching the khalti url

4.2.3 Special Requirements

Special Requirements
● Internet connection is required for payment and Khalti api integration. ● The amount entered by the user should matched the packages costs. ● The Khalti sandbox Api most works for the test Api key to works.

4.2.4 Pre-conditions

• User must be logged in to TrekNepal.

• User must open the TrekNepal Home / Packages page.

• Available trekking packages must be loaded successfully.

• Khalti (or selected payment gateway) API must be available and functioning.

• Internet connection must be active

4.2.5 Post-conditions

• User can successfully view selected package details.

• User can proceed to the payment gateway for booking confirmation.

• Booking is confirmed only after successful payment.

• System records and stores the booking details.

• User may receive booking confirmation.

4.2.6 Keyword
Booking System

4.2.7 Sample User Interface

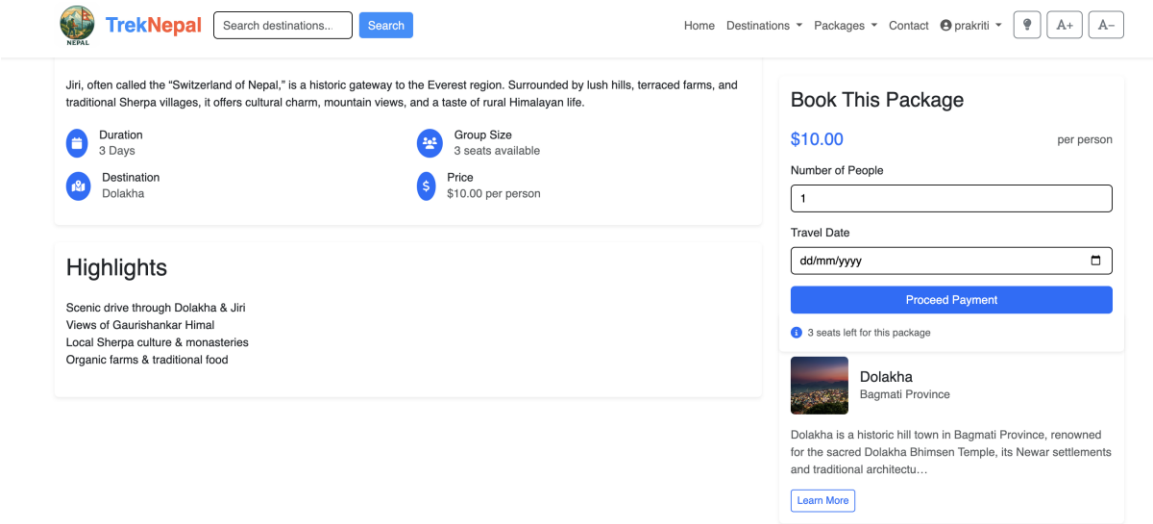

Figure 8:Sample User Interface of Booking System

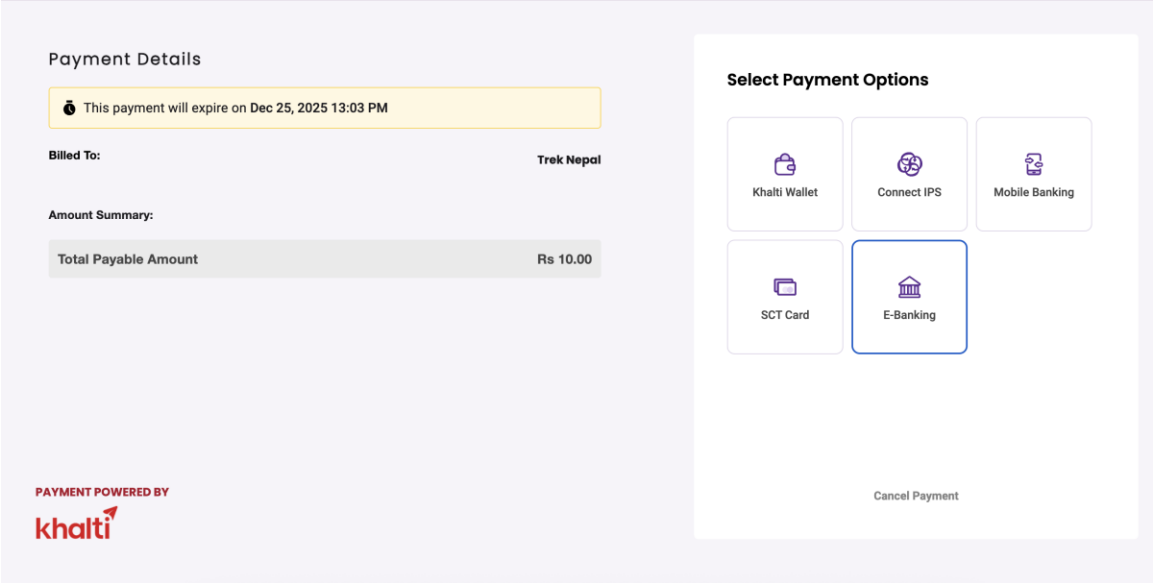

Figure 9:Sample User Interface of Booking System

4.2.8 Property Sheet

S.N	Field Name	Basic Type	Data Type	Field Size	Validation?	Enable?
1	Package List Display	UI Component	–	–	Must load packages successfully	Yes
2	Package Selection	Selection Component	String	–	Must select a valid package	Yes
3	Number of Participants Input	Text Input	Integer	–	Must be numeric & greater than 0	Yes
4	Date Selection	Date Picker	Date	–	Must select a valid date	Yes
5	Proceed to Payment Button	Button	–	–	Enabled only after required fields are filled	Yes
6	Payment Amount Input	Text / Numeric Input	Float / Integer	–	Must match package price	Yes
7	Payment Gateway	System Action	–	–	Must load Khalti payment page	Yes

S.N	Field Name	Basic Type	Data Type	Field Size	Validation?	Enable?
	Redirection				successfully	
8	Payment Confirmation Status	System Message	Boolean / String	–	Must confirm successful transaction	Yes

4.2.9 Process Flow Diagram

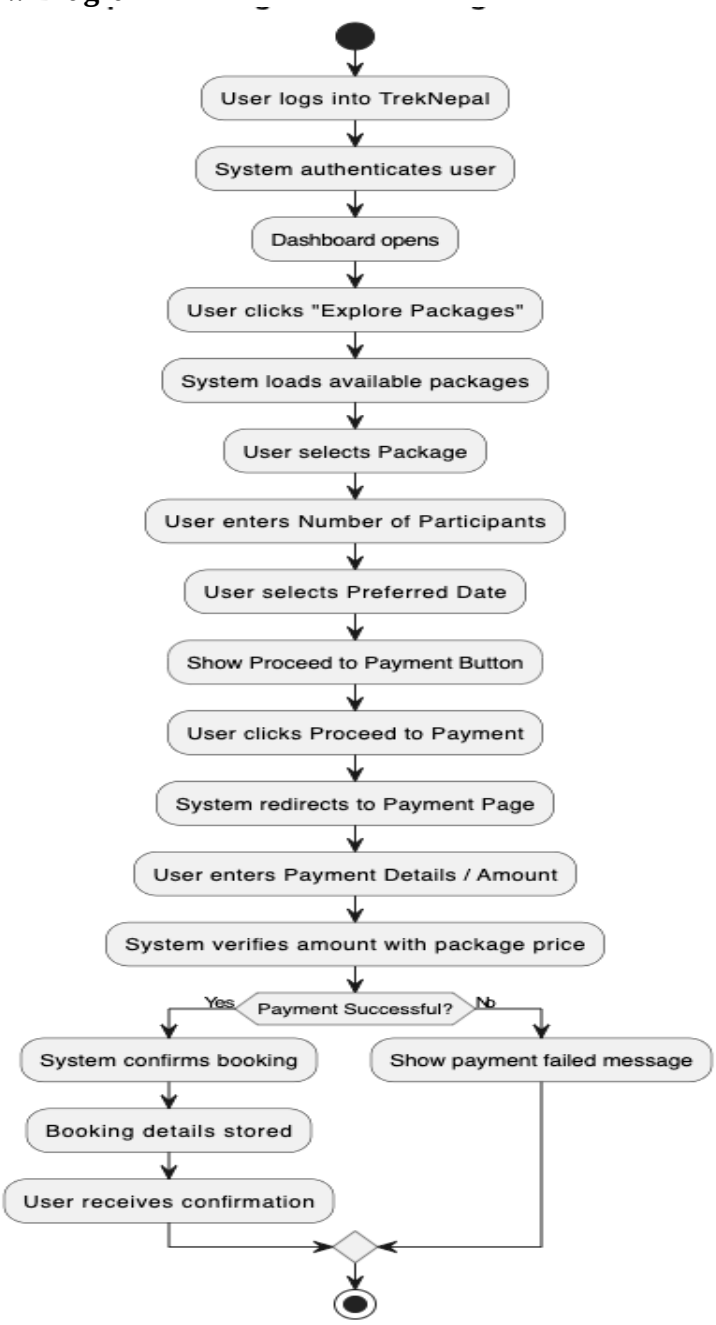

Figure 10:Interactive Map Process Flow

4.2.10 Use-Case Diagram

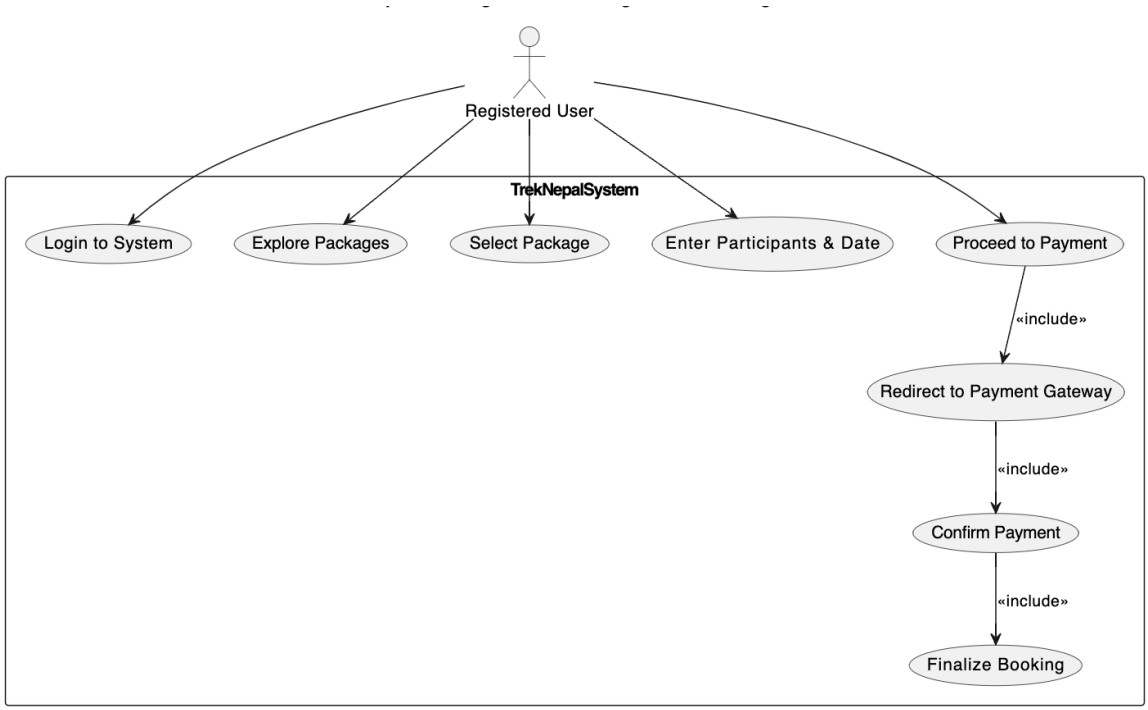

Figure 11:Use case diagram for Booking System

4.3 Use Case: Weather Dashboard

4.3.1 Use Case Details

This use case explains how the weather dashboard operates within the TrekNepal platform. After logging in, the user navigates to the home page and clicks the Weather button, which redirects them to the Weather Dashboard. The user enters the desired city name, and the system retrieves real-time weather information using the OpenWeatherMap API. The dashboard then displays the current weather conditions for the selected city, enabling travelers to review climate conditions and make informed decisions before planning their trip. (Ltd, 2024)

Actor(s)

Registered User (Logged-in User)

4.3.2 Detailed Process Flow

When the user opens the Home page, this use case begins.

Interaction	Response
User logs into TrekNepal system	System authenticates the user and opens the dashboard
User clicks on the **Weather** button	System redirects the user to the Weather Dashboard
User enters the **City Name**	System validates the input
User clicks the **Get Weather / Search** button	System sends request to OpenWeatherMap API
—	System fetches real-time weather data
—	System displays weather details for the entered city (temperature, condition, humidity, etc.)

4.3.3 Special Requirements

Special Requirements
• Internet connection is required to fetch weather data. • OpenWeatherMap API must be available and functioning • User must enter a valid city name. • System must handle invalid or unavailable city inputs gracefully.

4.3.4 Pre-conditions

• User must be logged in to TrekNepal.

• User must open the TrekNepal Home page.

• Weather Dashboard page must be accessible.

• OpenWeatherMap API key must be configured and active.

• Internet connection must be available.

4.3.5 Post-conditions

• User can view weather details of the entered city.

• System displays temperature, conditions, and other weather parameters.

• User can decide travel plans based on weather conditions.

• System may retain last searched city.

4.3.6 Keyword

Weather

4.3.7 Sample User Interface

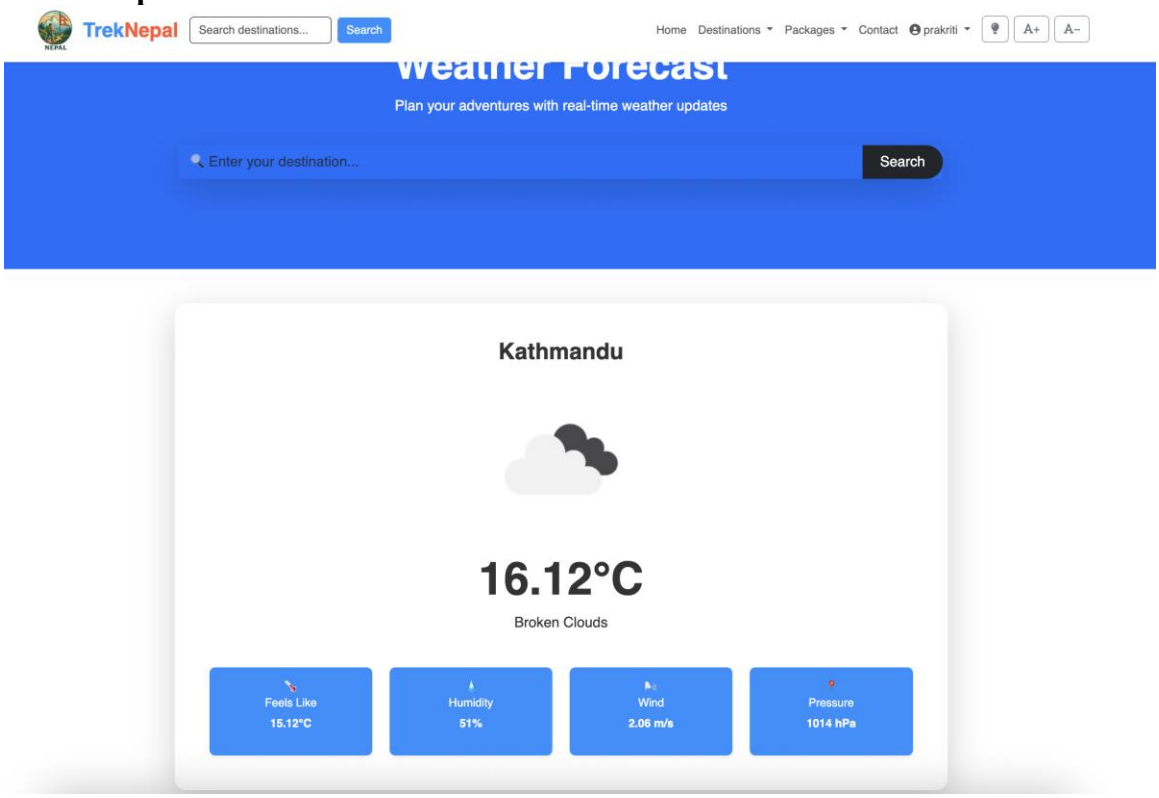

Figure 12:Sample User Interface of Weather Dashboard

4.3.8 Property Sheet

S.N	Field Name	Basic Type	Data Type	Field Size	Validation?	Enable?
1	City Input Field	Text Input	String	–	Must enter a valid city name	Yes
2	Get Weather / Search Button	Button	–	–	Enabled only after city is entered	Yes
3	Weather API Request	System Action	–	–	OpenWeatherMap API must respond	Yes
4	Weather Data Display	UI Component	–	–	Must display weather details correctly	Yes
5	Temperature Display	Text / Label	Float / String	–	Valid numeric temperature	Yes
6	Weather Condition Display	Text / Label	String	–	Must show correct status (e.g., Clear)	Yes
7	Humidity / Wind Info	Text / Label	Integer/String	–	Must show valid weather metrics	Yes
8	Error Message Display	System Message	String	–	Shows for invalid city or API failure	Yes

4.3.9 Process Flow Diagram

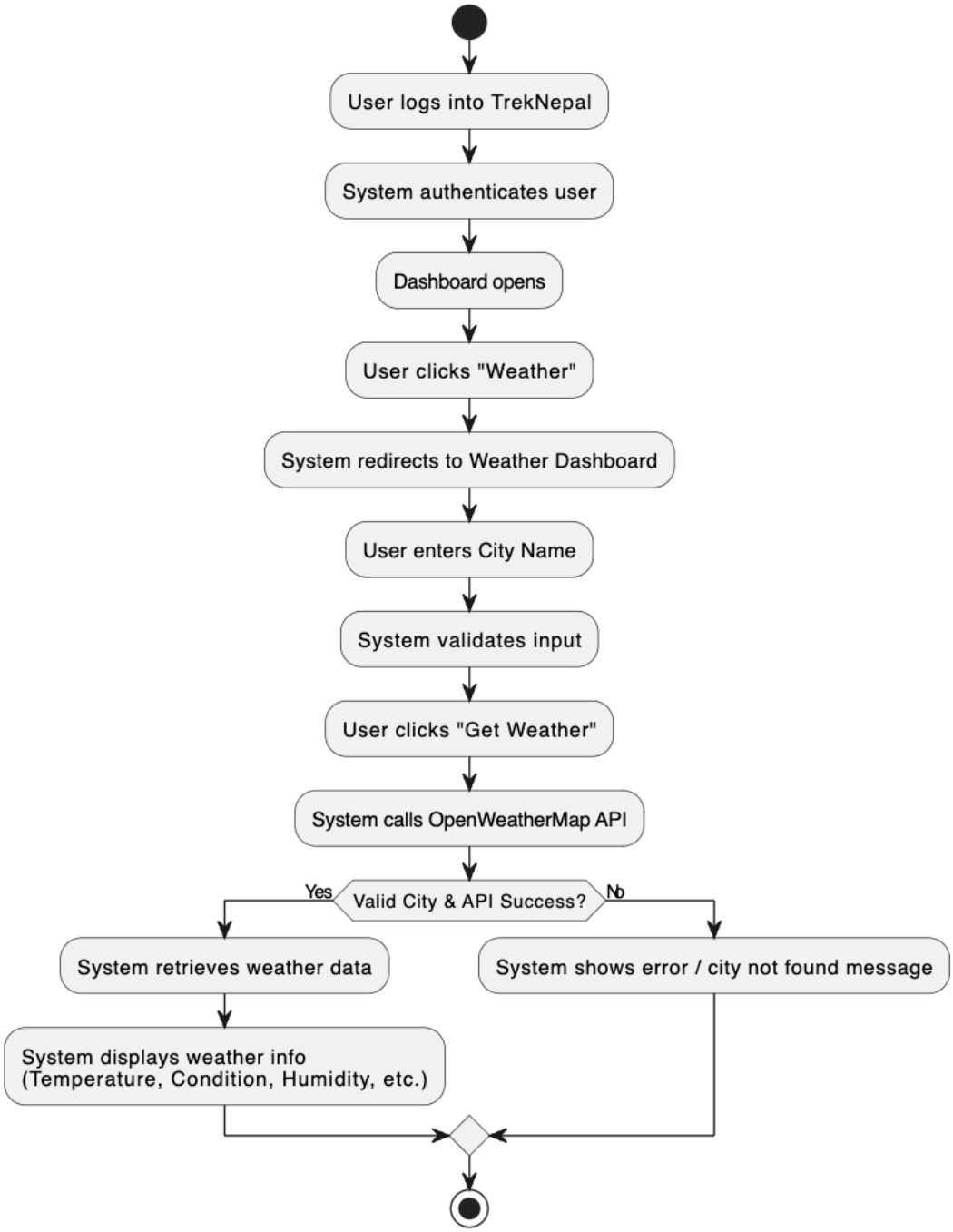

Figure 13:Weather Dashboard Process Flow

4.3.10 Use-Case Diagram

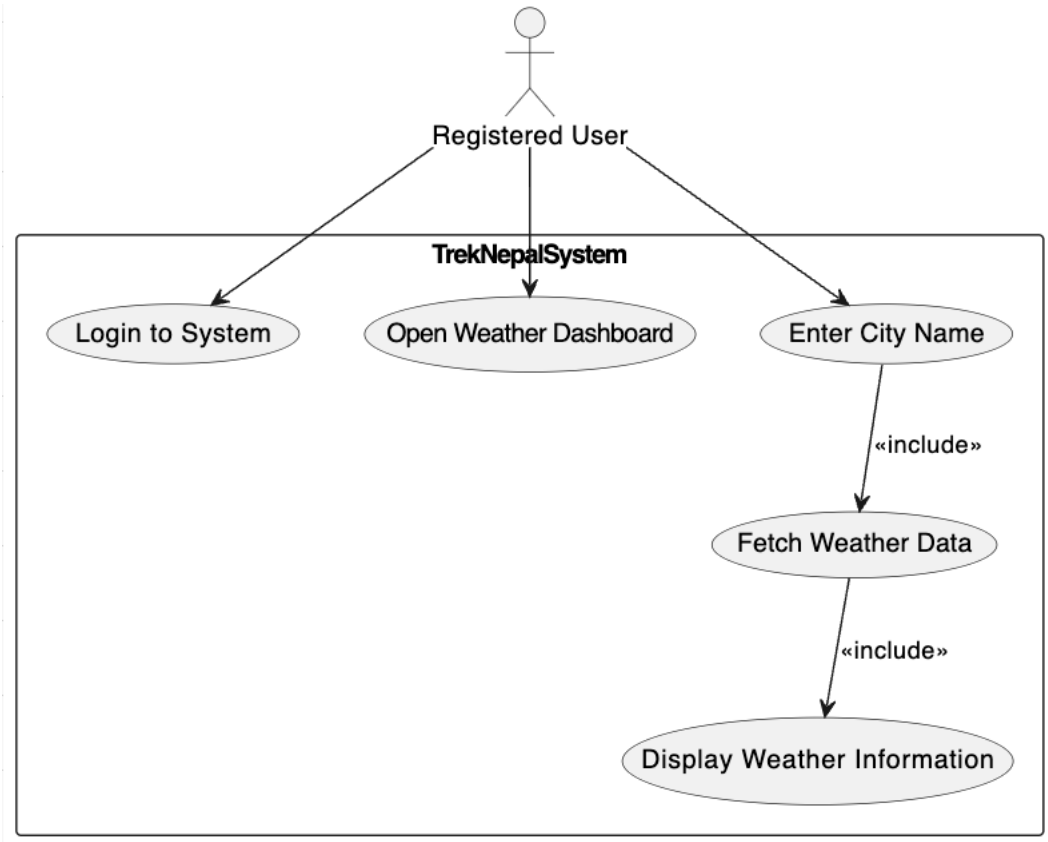

Figure 14:Use case diagram for Weather Dashboard

4.4 Use Case: Currency Converter

4.4.1 Use Case Details

This use case explains how the Currency Converter dashboard operates within the TrekNepal platform. After logging in, the user navigates to the home page and clicks the Currency Converter button, which redirects them to the Currency Converter Dashboard. The user selects the **From** and **To** currencies from dropdown menus, enters the amount, and the system retrieves the conversion rate using the relevant currency API. (ExchangeRate-API, 2024) The dashboard then displays the converted amount, enabling travelers to check currency rates before planning expenses for their trip.

Actor(s)

Registered User (Logged-in User)

4.4.2 Detailed Process Flow

When the user opens the Home page, this use case begins.

Interaction	Response
User logs into TrekNepal system	System authenticates the user and opens the

Interaction	Response
	dashboard
User clicks on the **Currency Converter** button	System redirects the user to the Currency Converter Dashboard
User selects **From** and **To** currencies and enters amount	System validates the inputs
User clicks the **Convert** button	System sends request to the currency API
—	System fetches real-time conversion rate
—	System displays converted amount for the selected currencies

4.4.3 Special Requirements

Special Requirements
• Internet connection is required to fetch currency data. • Currency API must be available and functioning. • User must select valid **From** and **To** currencies. • Amount entered must be numeric and positive. • System must handle API errors or invalid inputs gracefully.

4.4.4 Pre-conditions

• User must be logged in to TrekNepal.

• User must open the TrekNepal Home page.

• Currency Converter Dashboard page must be accessible.

• Currency API key must be configured and active.

• Internet connection must be available.

4.4.5 Post-conditions

• User can view the converted currency amount.

• System displays the current exchange rate used for conversion.

• User can plan expenses or budgets based on accurate currency data.

• System may retain last converted currencies.

4.4.6 Keyword

Currency Converter

4.4.7 Sample User Interface

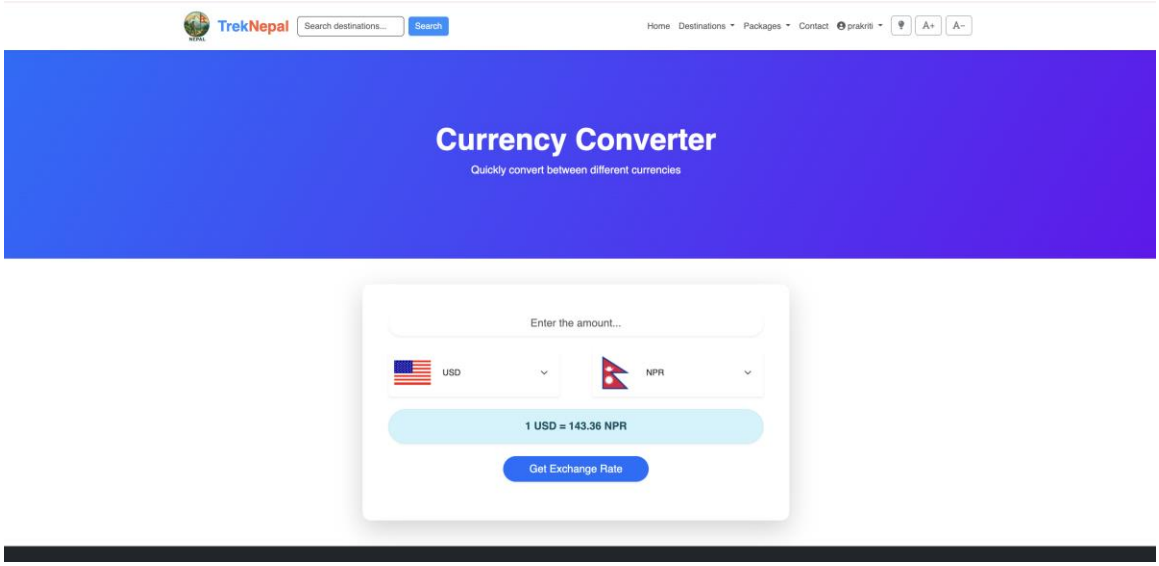

Figure 15:Sample User Interface of Currency Converter

4.4.8 Property Sheet

S.N	Field Name	Basic Type	Data Type	Field Size	Validation?	Enable?
1	From Currency Dropdown	Selection Input	String	–	Must select a valid currency	Yes
2	To Currency Dropdown	Selection Input	String	–	Must select a valid currency	Yes
3	Amount Input Field	Text Input	Float / Integer	–	Must enter numeric value > 0	Yes
4	Convert Button	Button	–	–	Enabled only after currencies & amount are entered	Yes
5	Currency API Request	System Action	–	–	Must fetch conversion rate successfully	Yes
6	Converted Amount Display	UI Component	Float / String	–	Must display conversion result correctly	Yes
7	Exchange Rate Display	Text / Label	Float / String	–	Must show correct exchange rate	Yes
8	Error Message Display	System Message	String	–	Shows for invalid input or API failure	Yes

4.4.9 Process Flow Diagram

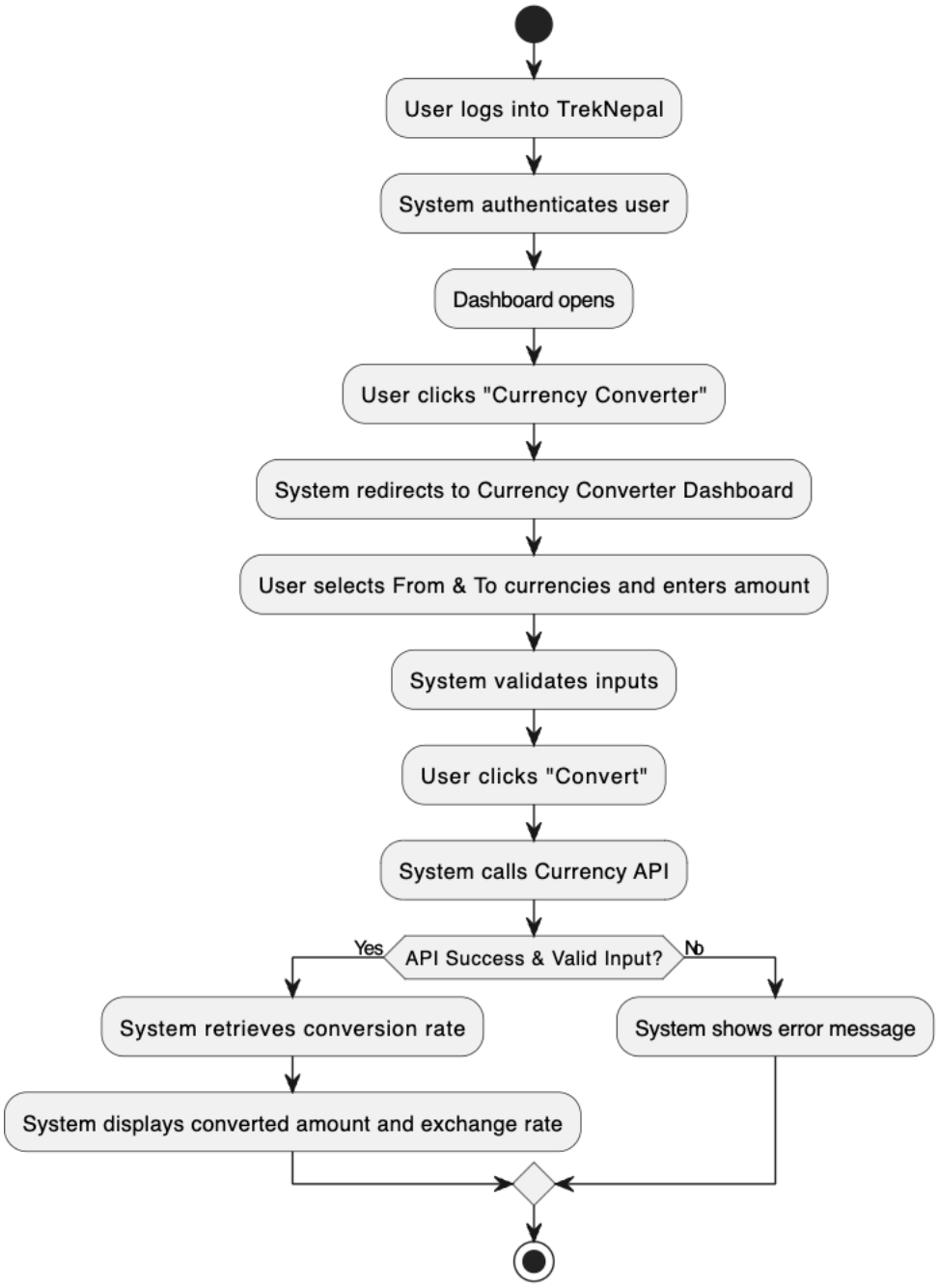

Figure 16:Currency Converter Process Flow

4.4.10 Use-Case Diagram

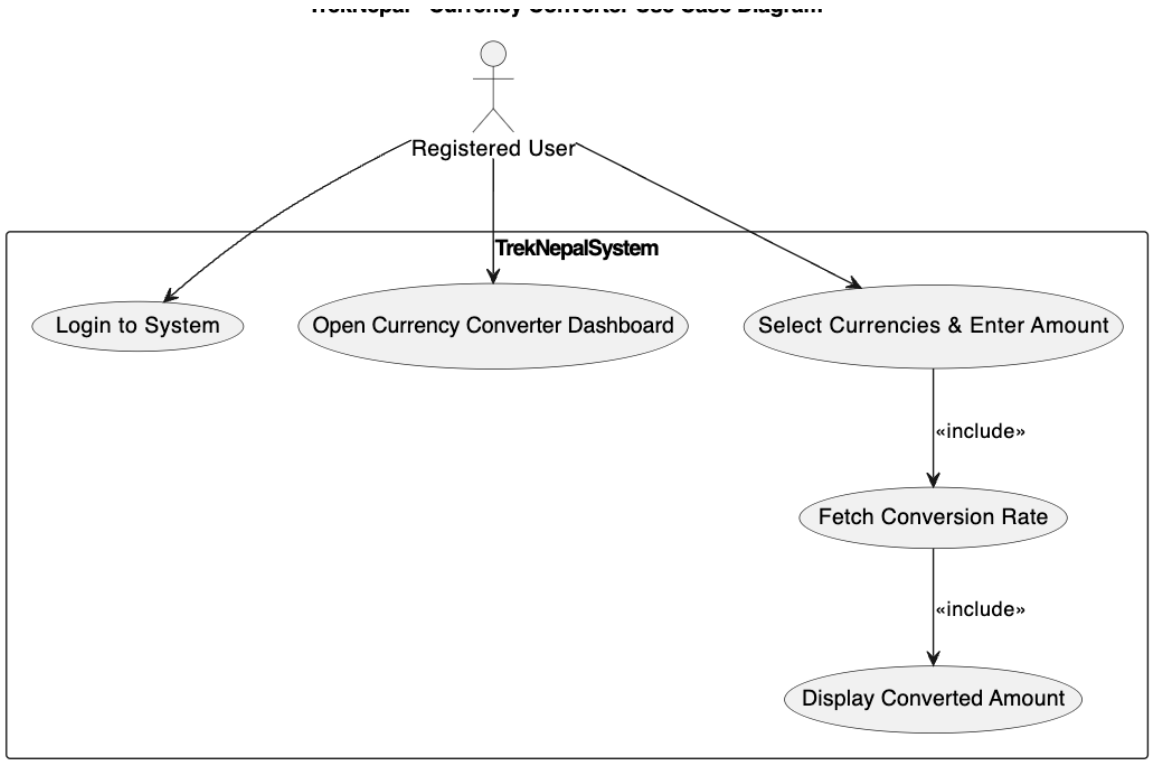

Figure 17:Use case diagram for Currency Converter

5. Database Design

Database will include:

• Users Table

• Trek Packages Tables

• Booking Table

• Reviews Table

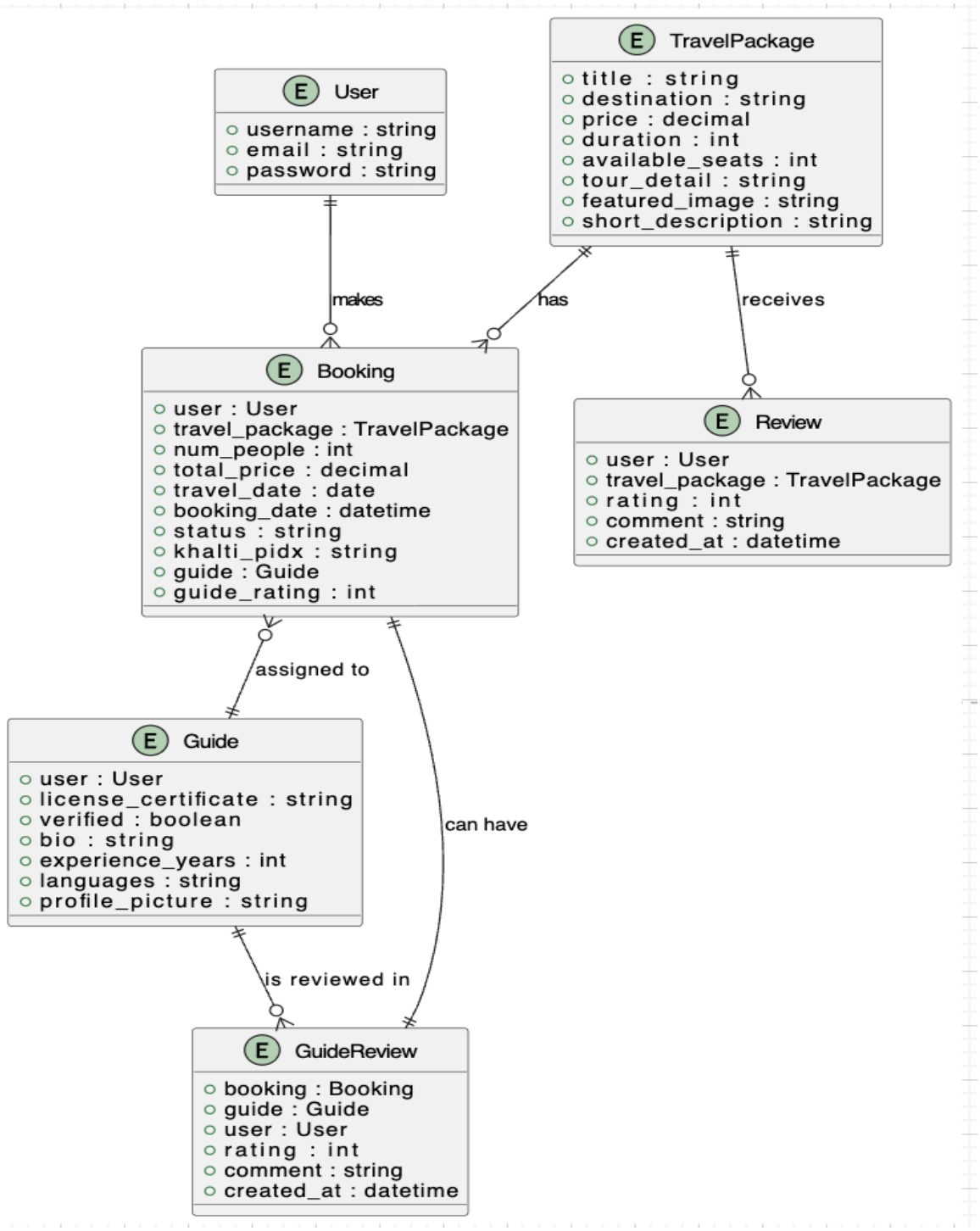

Figure 18:Entity-Relationship (ER) Diagram

6. Technology Stack

Frontend: HTML, CSS, JavaScript

Backend: Django

The backend of TrekNepal is developed using Django, a robust and scalable web framework suitable for rapid development. (Foundation, 2024)

Database: MySQL / dB SQLite

The application stores all user, trek, and booking data in a relational database managed using MySQL. (Corporation, 2024)

Hosting: Local Server

Other Tools: Payment Gateway API, Currency Converter API, OpenWeatherMap API

7. User Interface Design

The TrekNepal user interface is designed to provide a seamless and intuitive experience for users. The design process started with low-fidelity wireframes and progressed to high-fidelity mockups to ensure usability and consistency with user needs.

7.1 Wireframes (Low-Fidelity Design)

Wireframes illustrate the layout, navigation flow, and key user interactions before final implementation. The following wireframes represent the initial planning of key pages in the TrekNepal web application:

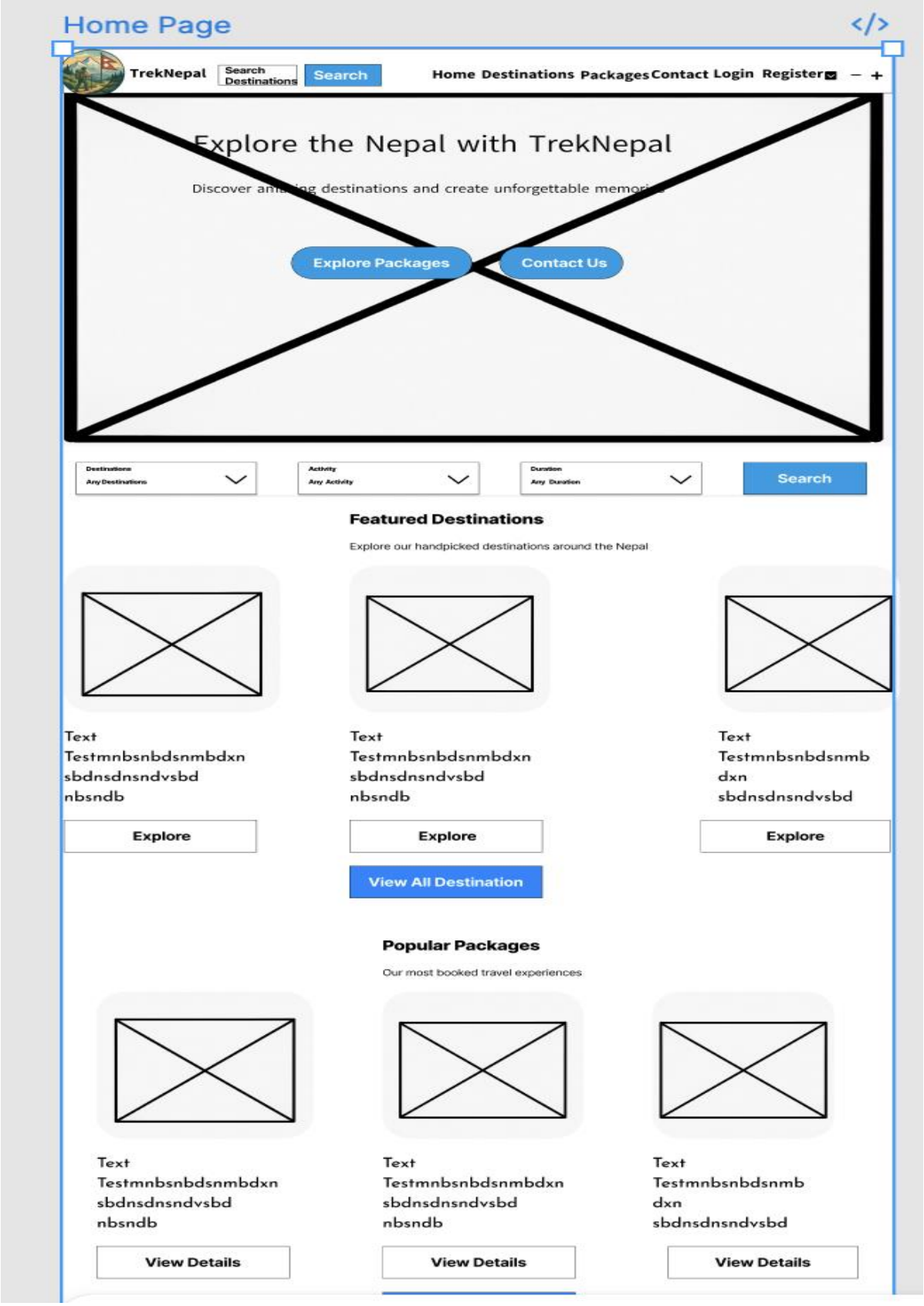
Home Page
TrekNepal
Search Destinations
Search
Home Destinations Packages Contact Login Register
Explore the Nepal with TrekNepal
Discover amazing destinations and create unforgettable memories
Explore Packages
Contact Us
Destinations
Any Destinations
Activity
Any Activity
Duration
Any Duration
Search
Featured Destinations
Explore our handpicked destinations around the Nepal
Text Testmnbsnbdsnmbdxn sbdnsdnsndvsbd nbsndb
Explore
Text Testmnbsnbdsnmbdxn sbdnsdnsndvsbd nbsndb
Explore
Text Testmnbsnbdsnmb dxn sbdnsdnsndvsbd
Explore
View All Destination
Popular Packages
Our most booked travel experiences
Text Testmnbsnbdsnmbdxn sbdnsdnsndvsbd nbsndb
View Details
Text Testmnbsnbdsnmbdxn sbdnsdnsndvsbd nbsndb
View Details
Text Testmnbsnbdsnmb dxn sbdnsdnsndvsbd
View Details

Figure 19:Low-Fidelity Design of Home Page

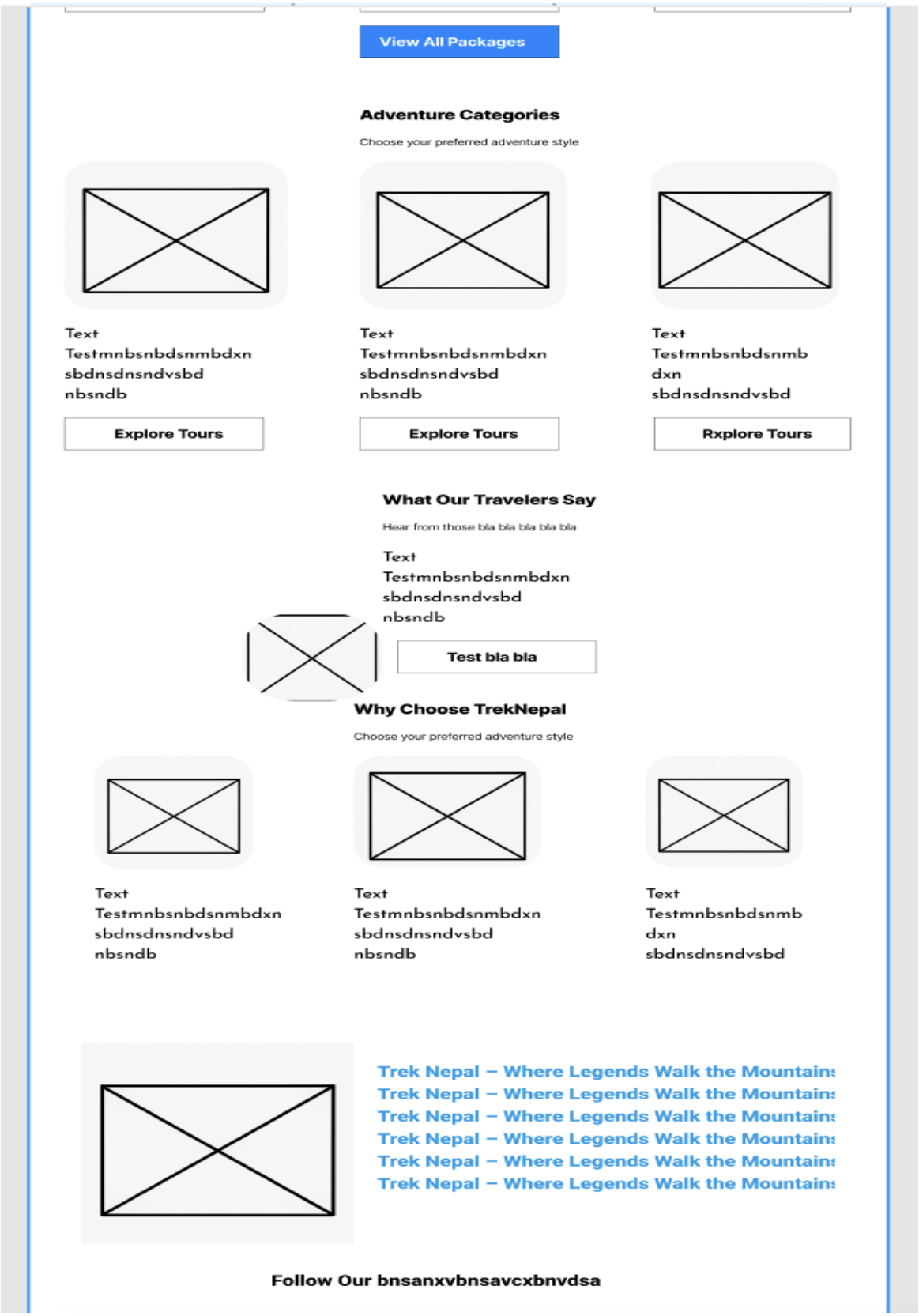

Figure 20:Low-Fidelity Design of Home Page

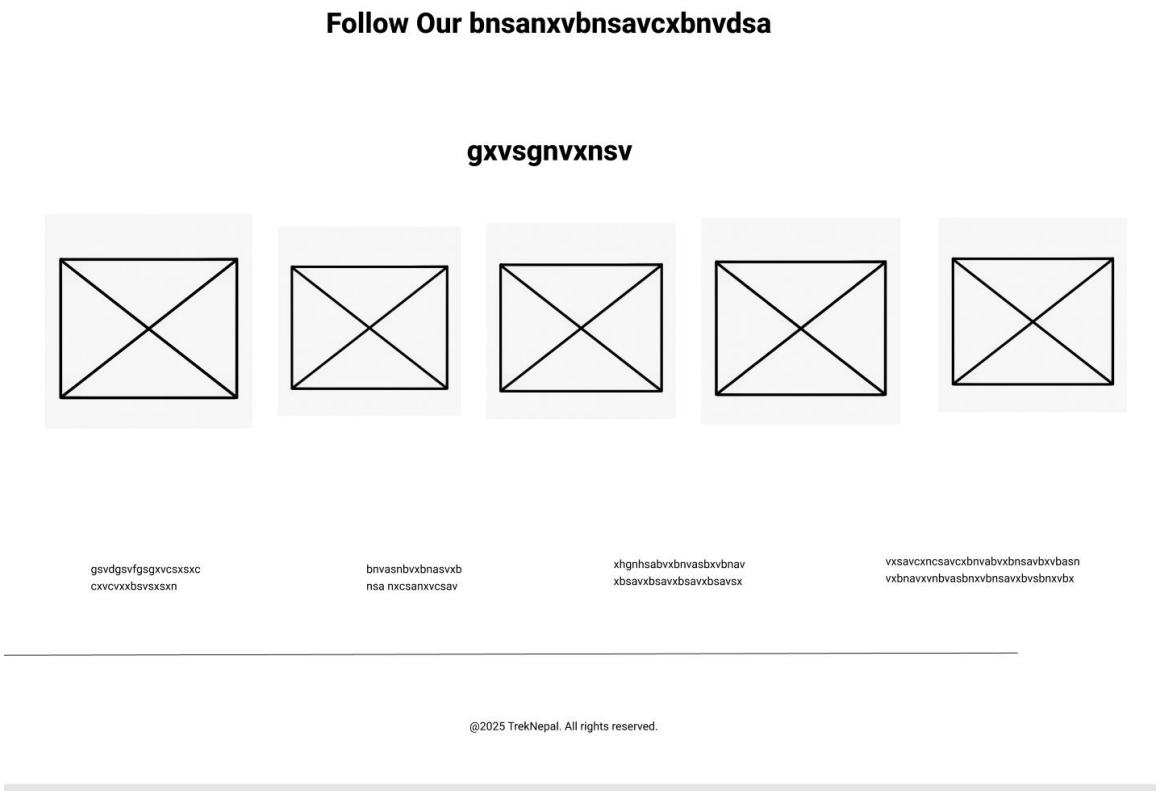

Figure 21:Low-Fidelity Design of Home Page

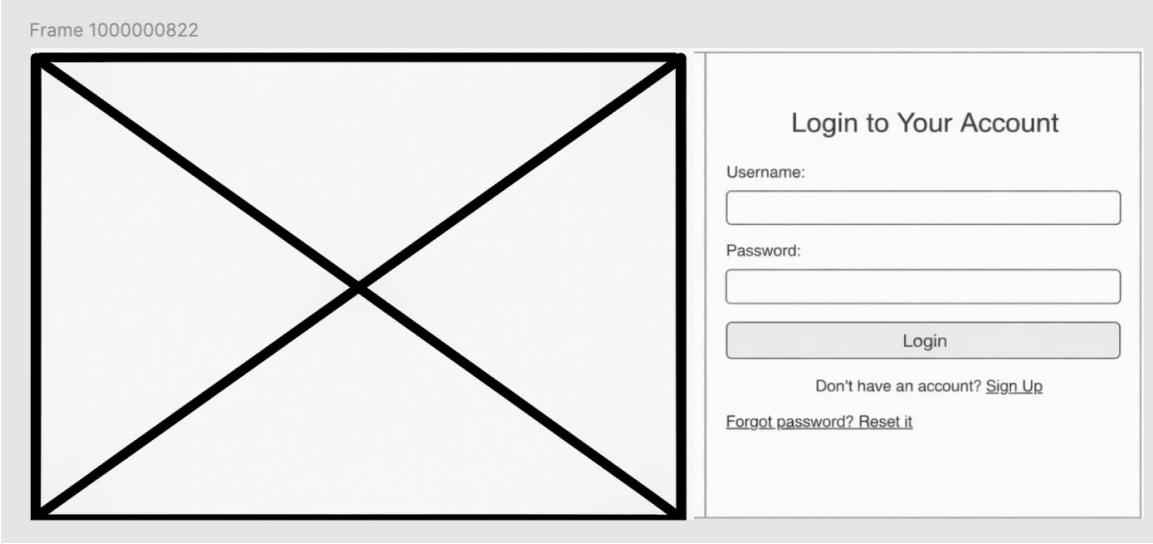

Figure 22: Low-fidelity wireframe of Login Page

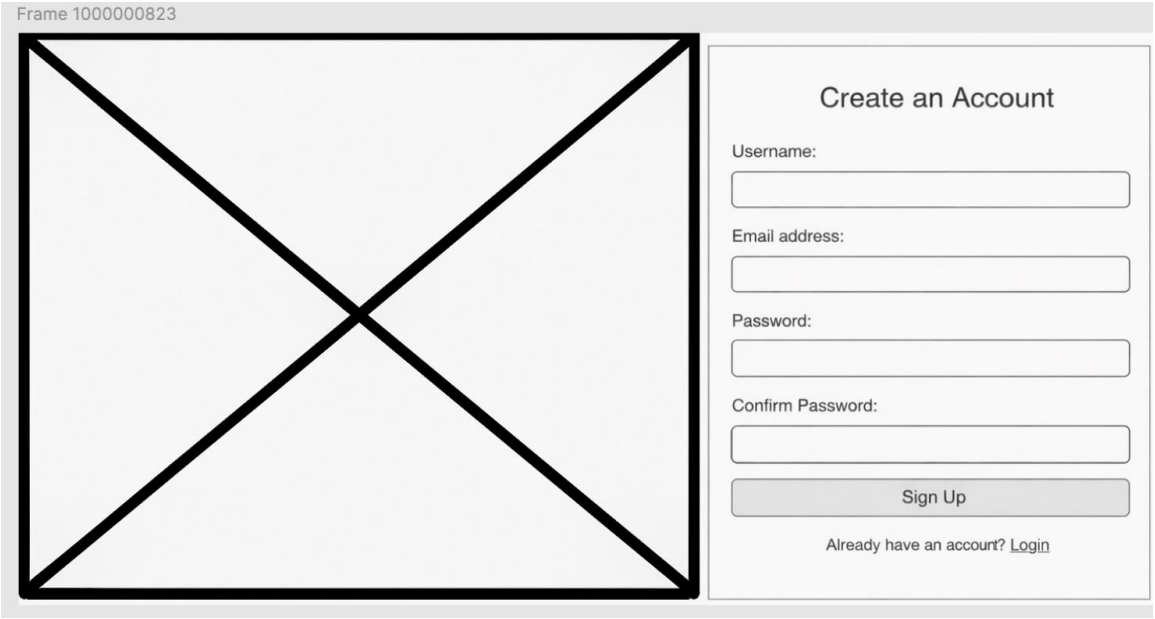

Figure 23:Low-fidelity wireframe of Registration Page

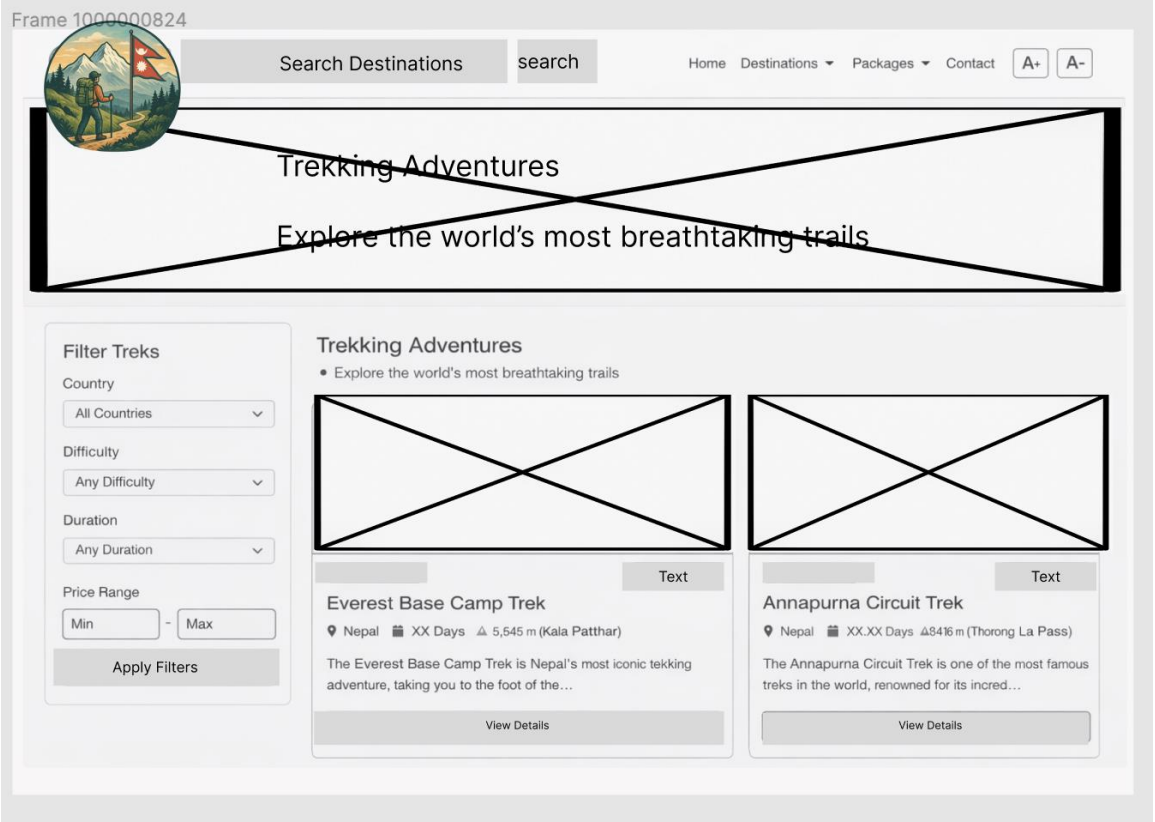

Figure 24:Low-fidelity wireframe of Trek Details Page

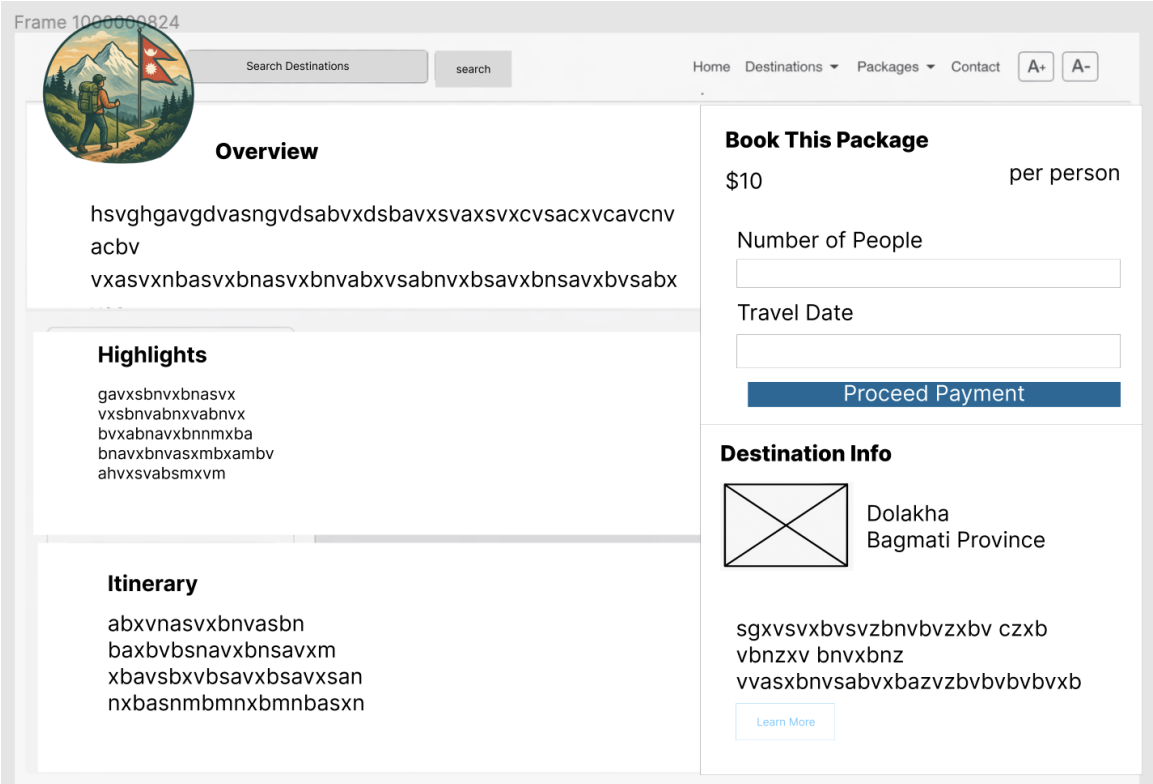

Figure 25:Low-fidelity wireframe of Booking Page

7.2 Implementation Comparison

To show that the implemented TrekNepal website follows the planned design, few screenshots of the live application are included, which display key pages, reflecting the layout and navigation of the wireframes, with visual adjustments for usability and responsiveness.

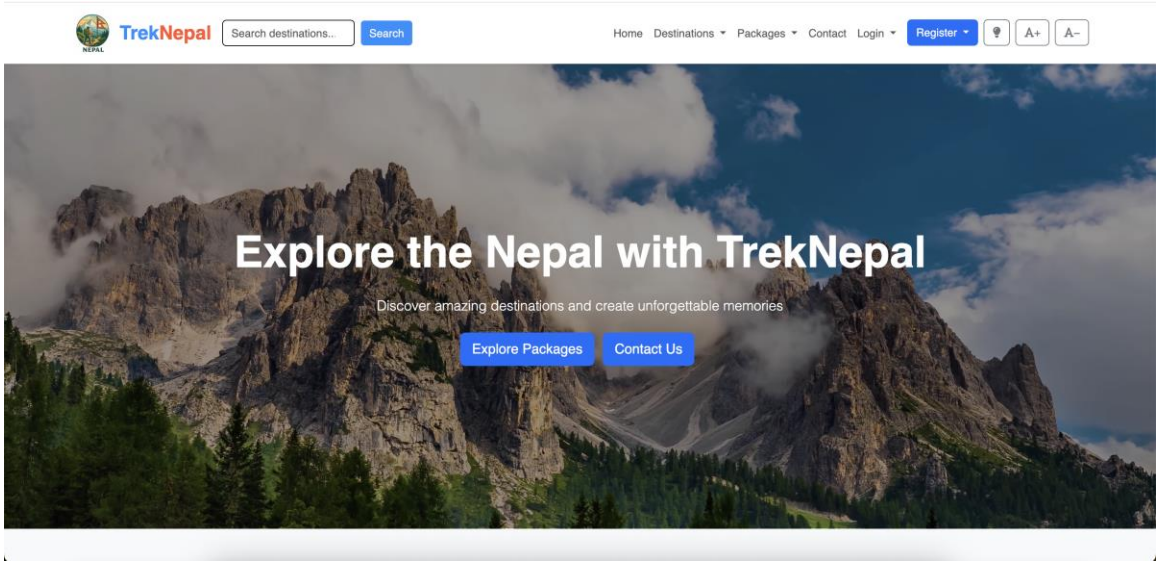

Figure 26:Implementation Comparison of Home Page with Lofi wireframes

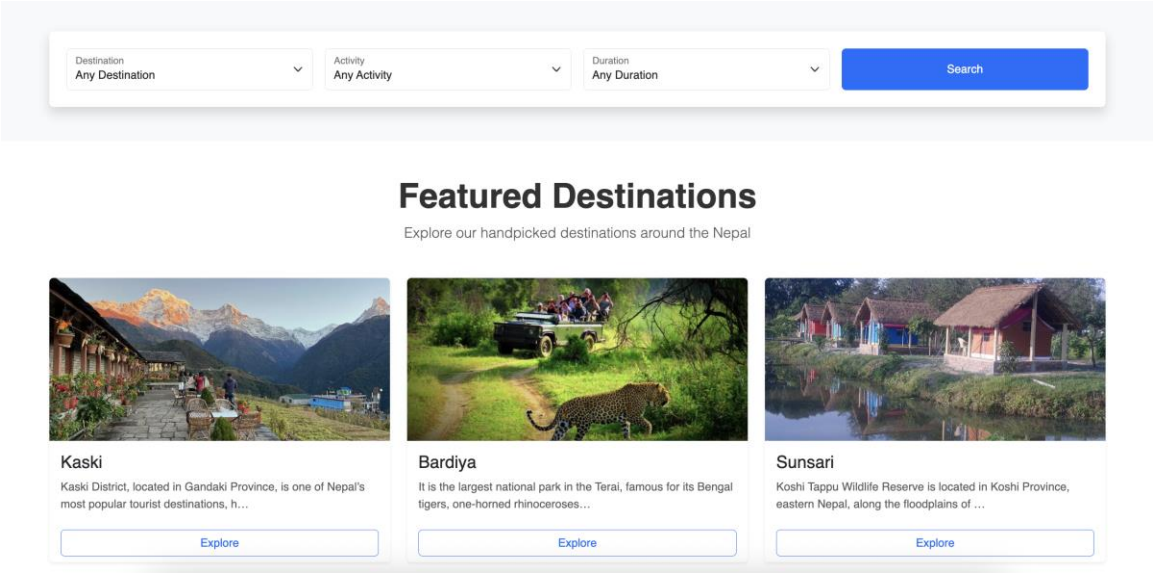

Figure 27:Implementation Comparison of Home Page with Lofi wireframes

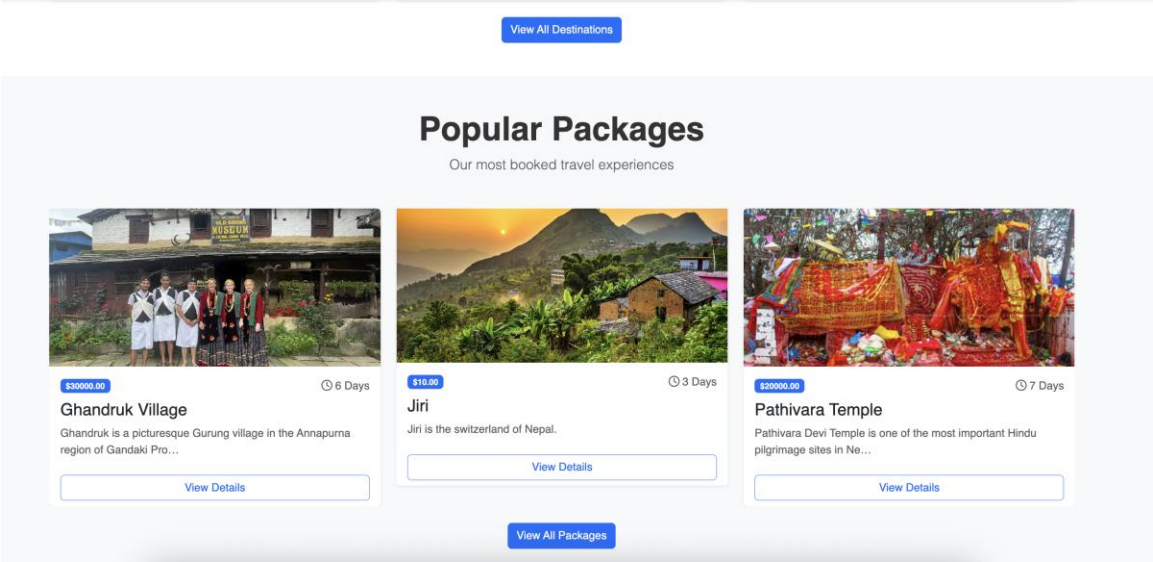

Figure 28:Implementation Comparison of Home Page with Lofi wireframes

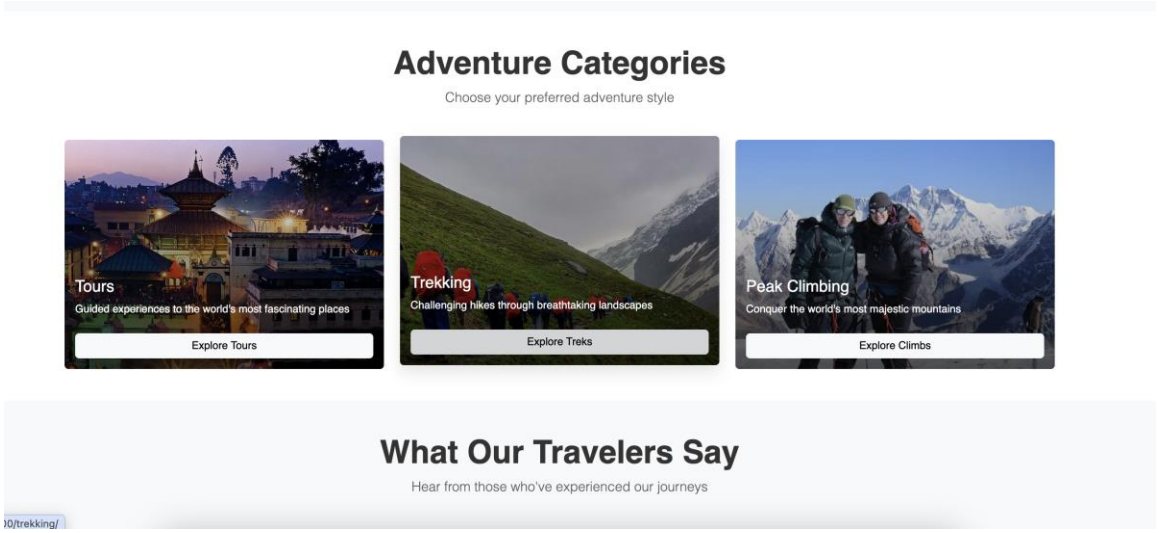

Figure 29:Implementation Comparison of Home Page with Lofi wireframes

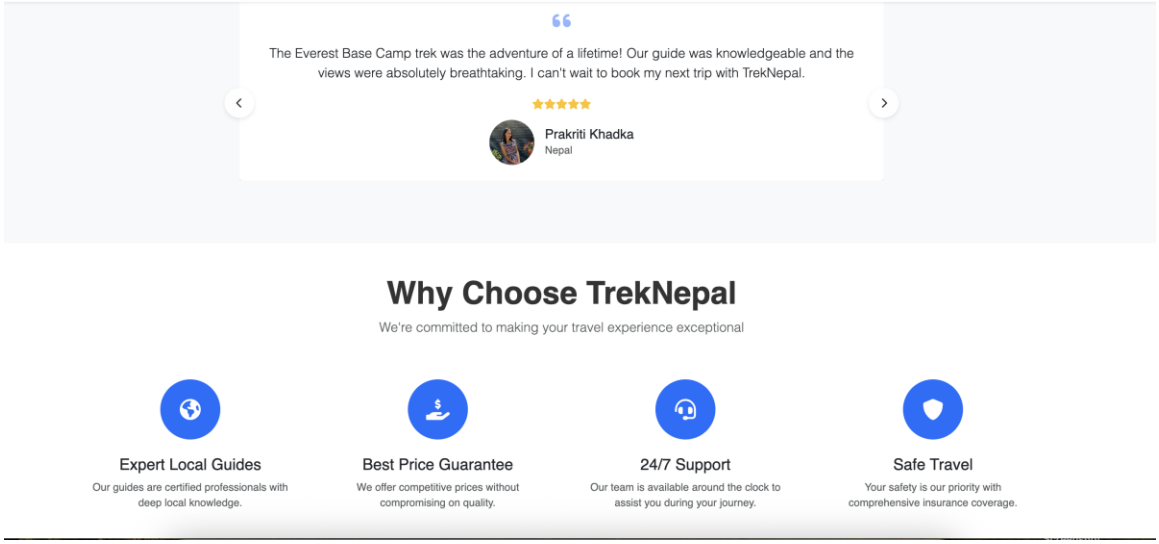

Figure 30:Implementation Comparison of Home Page with Lofi wireframes

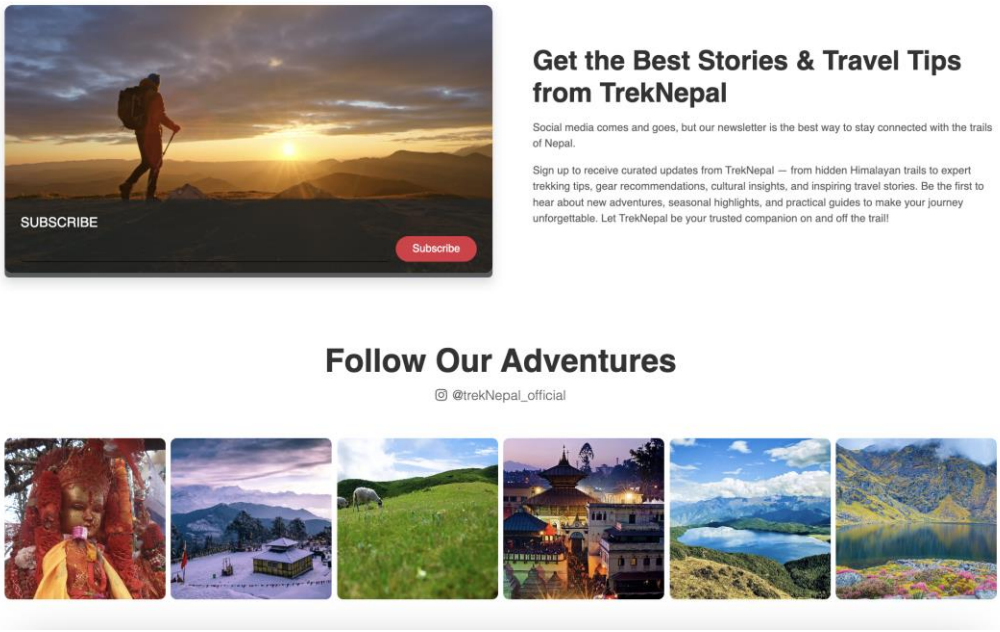

Figure 31:Implementation Comparison of Home Page with Lofi wireframes

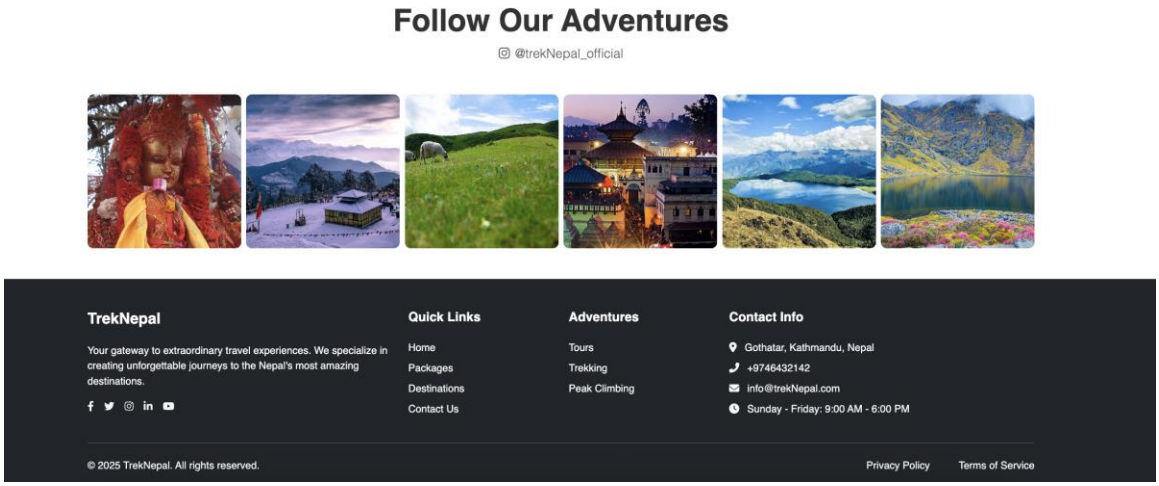

Figure 32:Implementation Comparison of Home Page with Lofi wireframes

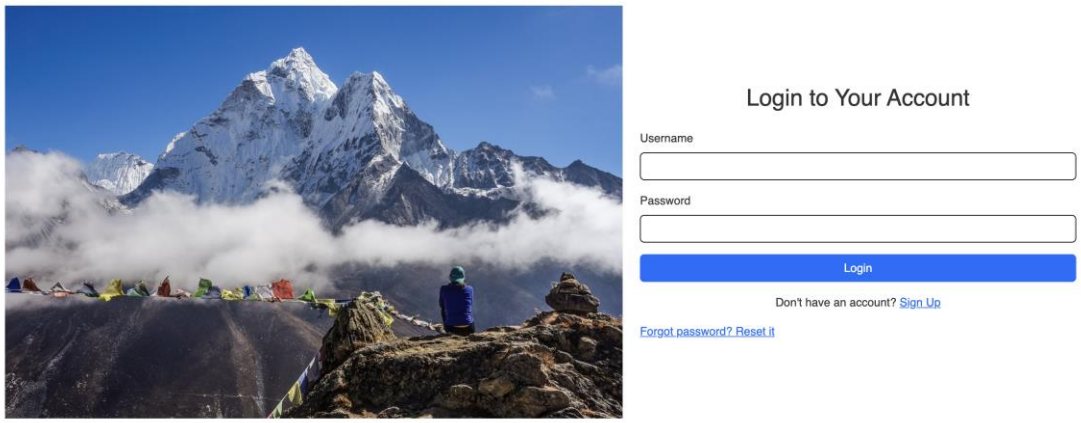

Figure 33:Implemented Login Page interface

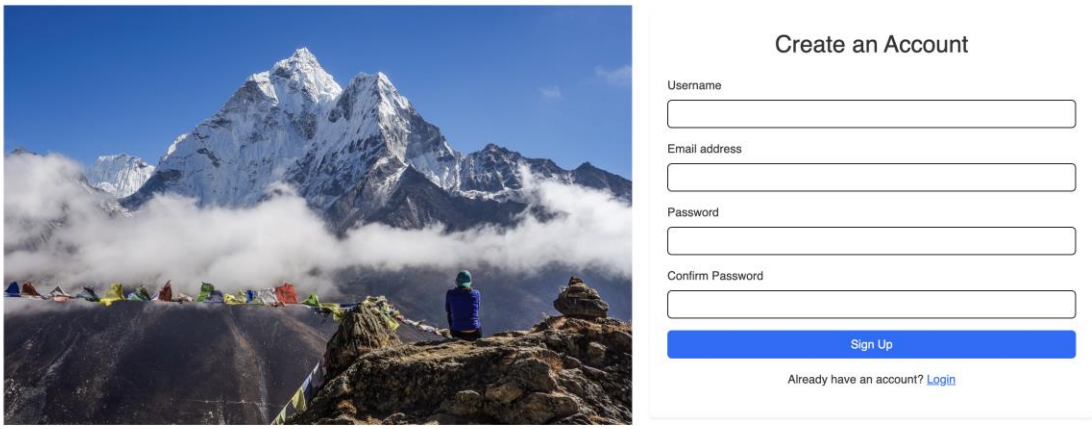

Figure 34:Implemented Registration Page interface

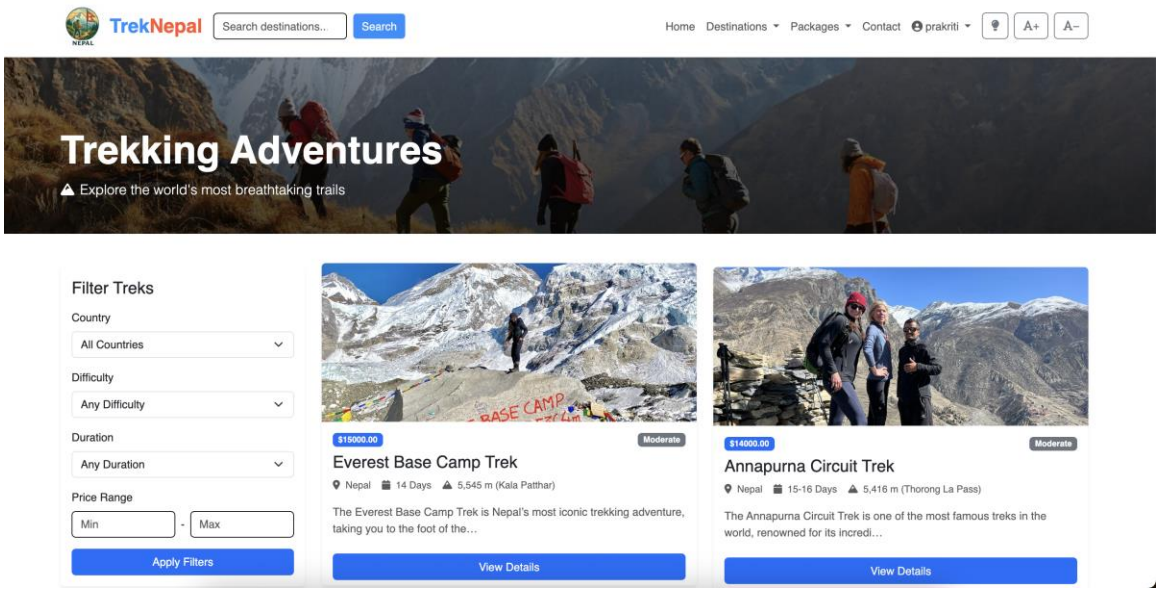

Figure 35:Implemented Trek Details Page interface

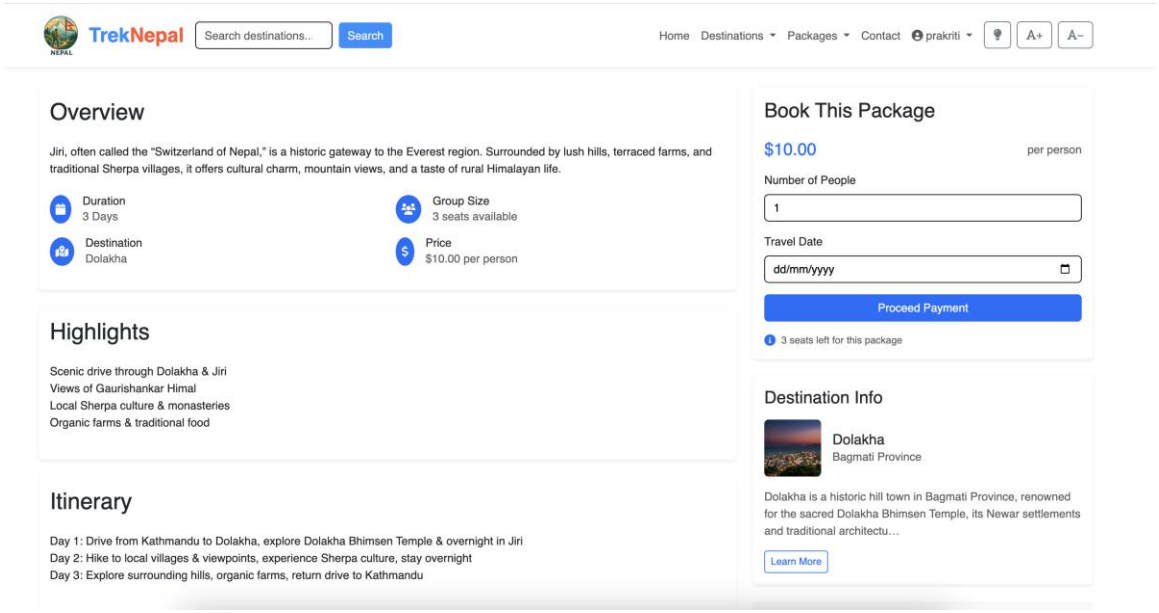

Figure 36:Implemented Booking Page interface

8. Security & Reliability

• Secure Authentication

• Encrypted Data Storage where needed

• Error Handling

• Session management, and input validation

9. Deployment Plan

• Host on web server or cloud

• Setup database server

• Deploy frontend

10. Test Plan and Evaluation

10.1 Testing Objectives

The objective of testing is to ensure that the TrekNepal web application functions correctly, is user-friendly, secure, and reliable for end users.

10.2 Test Environment

- Browser: Google Chrome
- Device: Laptop
- Server: Localhost / Cloud Server
- Database: MySQL / dB SQLite

10.3 Types of Testing

Functional Testing

- User registration and login
- Trek package browsing
- Booking and payment process
- Admin management features

Performance Testing

- Page loading speed
- System response under users

Usability Testing

- Ease of navigation
- User interface clarity

Security Testing

- Authentication and authorization
- Data protection and secure access

10.4 Sample Test Cases

Test Case ID	Description	Expected Result
TC01	User Registration	Account created successfully
TC02	User Login	User redirected to dashboard
TC03	Trek Booking	Booking confirmed
TC04	Admin Login	Admin dashboard accessible

10.5 Evaluation Summary

The system was evaluated based on functionality, usability, and performance. Test results indicate that the TrekNepal application meets the required objectives and provides a reliable platform for trekkers.

11. Conclusion

The TrekNepal artifact successfully demonstrates a complete web-based trekking management system. The design and testing ensure usability, reliability, and scalability.

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